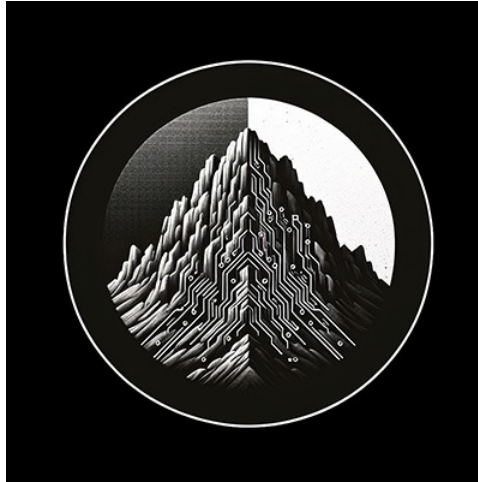




RTL Design Sherpa

RTL AMBA AXI4-Lite Module Reference — Rev 0.90

July 2026

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1 AXIL4 (AXI4-Lite) Modules

Location: rtl/amba/axil4/ **Test Location:** val/amba/ **Status:** ✓ Production Ready

1.1 Overview

The AXIL4 subsystem provides a complete implementation of the ARM AMBA AXI4-Lite protocol—a simplified subset of AXI4 designed for control register access with reduced signal count and single-beat transactions.

1.1.1 Key Features

- ✓ **AXI4-Lite Protocol:** Simplified subset (no burst, no ID, no QoS)
- ✓ **Single-Beat Transactions:** Always 1 transfer per transaction
- ✓ **Reduced Signals:** 40-50% fewer signals vs full AXI4
- ✓ **Buffered Interfaces:** Elastic buffering via gaxi_skid_buffer
- ✓ **Monitoring Infrastructure:** Integrated transaction monitoring
- ✓ **Clock Gating Variants:** Power-optimized versions
- ✓ **Typical Use:** Control registers, CSRs, peripheral access

1.2 Module Categories

1.2.1 Core Master/Slave Modules

Module	Description	Documentation	Status
axil4_master_rd	AXIL4 read master with buffered channels	axil4_master_rd.md	✓ Documented
axil4_master_wr	AXIL4 write master with buffered channels	axil4_master_wr.md	✓ Documented
axil4_slave_rd	AXIL4 read slave with buffered channels	axil4_slave_rd.md	✓ Documented
axil4_slave_wr	AXIL4 write slave with buffered channels	axil4_slave_wr.md	✓ Documented

1.2.2 Monitor Modules (Verification)

Monitor wrappers live in `rtl/amba/monitor/` (not `rtl/amba/axil4/`). They are dedicated AXIL4 wrappers — there is no IS_AXI-style parameter overload of the AXI4 wrappers — and they share `axi_monitor_base` and the 128-bit packet format with the AXI4 wrappers.

Module	Description	Documentation	Status
axil4_master_rd_mon	Read master with integrated monitoring	axil4_master_rd_mon.md	✓ Documented
axil4_master_wr_mon	Write master with integrated monitoring	axil4_master_wr_mon.md	✓ Documented
axil4_slave_rd_mon	Read slave with integrated monitoring	axil4_slave_rd_mon.md	✓ Documented
axil4_slave_wr_mon	Write slave with integrated monitoring	axil4_slave_wr_mon.md	✓ Documented

1.2.3 Clock-Gated Variants

All clock-gated variants documented in: [axil4_clock_gating_guide.md](#)

Module	Base Module	Status
axil4_master_rd_cg	axil4_master_rd	✓ Documented
axil4_master_wr_cg	axil4_master_wr	✓ Documented
axil4_slave_rd_cg	axil4_slave_rd	✓ Documented
axil4_slave_wr_cg	axil4_slave_wr	✓ Documented
axil4_master_rd_mon_cg	axil4_master_rd_mon	✓ Documented
axil4_master_wr_mon_cg	axil4_master_wr_mon	✓ Documented
axil4_slave_rd_mon_cg	axil4_slave_rd_mon	✓ Documented
axil4_slave_wr_mon_cg	axil4_slave_wr_mon	✓ Documented

1.3 AXI4-Lite Protocol Overview

1.3.1 Simplified vs Full AXI4

Feature	AXI4	AXI4-Lite
Burst Support	Yes (1-256 beats)	No (always 1 beat)
ID Signals	ARID/AWID/RID/BID	None
Out-of-Order	Supported via ID	Not supported
Data Width	8-1024 bits	32 or 64 bits
QoS/Region	Supported	Not supported
Lock/Cache	Full support	None (no AxLOCK, no AxCACHE)
Signal Count	~40-50 signals	~20-25 signals
Typical Use	High-bandwidth data	Control registers

1.3.2 Channel Architecture

AXI4-Lite defines five independent channels. Any one transaction uses only a subset of them: a write uses three (AW, W, B) and a read uses two (AR, R). The read and write channel sets are fully independent and can be in flight concurrently.

Write Transaction (3 channels): 1. **AW (Write Address)** - Address and protection 2. **W (Write Data)** - Data and byte strobes 3. **B (Write Response)** - Completion status

Read Transaction (2 channels): 1. **AR (Read Address)** - Address and protection 2. **R (Read Data)** - Data and response

1.3.3 Key Signals (Simplified)

Address Channels (AR/AW): - ARADDR/AWADDR - Byte address - ARPROT/AWPROT - Protection attributes (3 bits) - ARVALID/AWVALID, ARREADY/AWREADY - Handshake

Data Channels (R/W): - RDATA/WDATA - Transfer data (32 or 64 bits) - RRESP/BRESP - Response (OKAY, SLVERR, DECERR) - WSTRB - Write strobes (byte lane enables) - RVALID/WVALID, RREADY/WREADY - Handshake

1.4 Quick Start

1.4.1 Basic Read Master

```
axil4_master_rd #(
    .AXIL_ADDR_WIDTH(32),
    .AXIL_DATA_WIDTH(32),
    .SKID_DEPTH_AR(2),    // 2 entries
    .SKID_DEPTH_R(4)      // 4 entries
) u_rd_master (
    .aclk                (axi_clk),
    .aresetn              (axi_resetn),

    // Frontend interface (from CPU)
    .fub_araddr           (cpu_araddr),
    .fub_arprot            (cpu_arprot),
    .fub_arvalid           (cpu_arvalid),
    .fub_arready           (cpu_arready),

    .fub_rdata             (cpu_rdata),
    .fub_rresp             (cpu_rresp),
    .fub_rvalid            (cpu_rvalid),
    .fub_rready            (cpu_rready),

    // Master interface (to interconnect)
    .m_axil_araddr         (periph_araddr),
    // ... rest of AR/R signals

    .busy                  (rd_busy)
);
```

1.4.2 Basic Write Slave

```
axil4_slave_wr #(
    .AXIL_ADDR_WIDTH(32),
    .AXIL_DATA_WIDTH(32),
    .SKID_DEPTH_AW(2),
    .SKID_DEPTH_W(2),
    .SKID_DEPTH_B(2)
) u_wr_slave (
    .aclk                (axi_clk),
    .aresetn              (axi_resetn),

    // Slave interface (from interconnect)
    .s_axil_awaddr        (interconnect_awaddr),
    // ... rest of AW/W/B signals
);
```

```

    // Backend interface (to register block)
    .fub_awaddr      (regfile_awaddr),
    // ... rest of AW/W/B signals

    .busy            (wr_busy)
);

1.4.3 Monitor Integration
axil4_master_rd_mon #(
    .AXIL_ADDR_WIDTH(32),
    .AXIL_DATA_WIDTH(32),
    .UNIT_ID(1),
    .AGENT_ID(10),
    .MAX_TRANSACTIONS(8),    // Reduced for AXIL (axi4*_mon defaults
to 16)
    .ENABLE_FILTERING(1)
) u_rd_mon (
    .aclk            (axi_clk),
    .aresetn         (axi_resetn),

    // AXIL interfaces
    .fub_axil_araddr (cpu_araddr),
    // ... rest of AR/R signals

    // Monitor configuration
    .cfg_monitor_enable    (1'b1),
    .cfg_error_enable      (1'b1),
    .cfg_timeout_enable    (1'b1),
    .cfg_perf_enable       (1'b0), // Avoid congestion
    .cfg_timeout_cycles    (16'd1000),

    // Filtering masks - see note below on polarity
    .cfg_axi_pkt_mask      (16'b1111_1111_1111_0110), // Keep Error
+ Timeout
    // ... rest of masks

    // Monitor bus output
    .monbus_valid          (mon_valid),
    .monbus_ready          (mon_ready),
    .monbus_packet         (mon_packet),
    .monbus_timestamp      (mon_timestamp),

    // Status
    .busy                  (rd_busy),
    .active_transactions   (rd_active),

```

```
        .error_count                (rd_errors)
    );
```

cfg_axi_pkt_mask polarity and indexing. The mask is 16 bits because it is indexed by the 4-bit packet_type field of the monitor packet — one mask bit per packet type — not by the 128-bit packet width. A **set bit drops** that packet type (pkt_drop = cfg_axi_pkt_mask[pkt_type] in axi_monitor_filtered). So 16'b1111_1111_1111_0110 clears bit 0 (PktTypeError) and bit 3 (PktTypeTimeout) to let those through and drops everything else. See the [Monitor Packet Specification](#) for the packet-type enum.

1.5 Testing

All AXIL4 modules are verified using CocoTB-based testbenches:

```
# Run all AXIL4 tests
```

```
pytest val/amba/test_axil4*.py -v
```

```
# Run specific module tests
```

```
pytest val/amba/test_axil4_master_rd.py -v
```

```
pytest val/amba/test_axil4_slave_wr.py -v
```

```
pytest val/amba/test_axil4_master_rd_mon.py -v
```

```
# Run with waveforms
```

```
pytest val/amba/test_axil4_master_rd.py --vcd=waves.vcd -v
```

1.6 Terminology

Term	Meaning
FUB	Functional Unit Block — the design-internal logic on the far side of the module from the external AXI4-Lite bus. FUB-side ports are prefixed fub_.
Skid buffer	A small elastic FIFO (gaxi_skid_buffer) placed in a valid/ready channel. It registers the handshake so neither side sees a combinational path to the other, at the cost of one clock of latency.
CG	Clock Gating. The _cg module suffix denotes the clock-gated variant.

Term	Meaning
MonBus	The 128-bit monitor packet bus (plus 64-bit side-band timestamp) emitted by the <code>_mon</code> wrappers.

1.7 Known Limitations

- **No clock domain crossing.** All modules are fully synchronous to `aclk`. Cross domains with `gaxi_skid_buffer_async` or a CDC FIFO outside these modules.
- **No address decode or protection enforcement.** The modules are transport only: `AxPROT` is carried through untouched and never checked. Address decode belongs to the interconnect or the register block.
- **No error injection or response rewriting.** `RRESP` / `BRESP` pass through from the backend unmodified.
- **No write-data reordering or AW/W matching.** The AW and W channels are buffered independently; nothing in these modules pairs an address with its data. A backend that requires AW before W must enforce that itself.
- **Unaligned and narrow accesses are the backend's problem.** AXI4-Lite has no `AxSIZE`; partial-word access is expressed only through `WSTRB`, which the backend must honor.
- **`gaxi_skid_buffer` constrains `SKID_DEPTH_*` to 2..8 inclusive.** Other values are not supported.
- **Clock gating is ASIC-oriented.** `clock_gate_ctrl` instantiates an ICG cell; on FPGA this does not map to true clock gating. There is no scan/test bypass port — set `cfg_cg_enable = 0` to disable gating.

1.8 Design Patterns

1.8.1 Pattern 1: Buffered Master/Slave

All core modules use `gaxi_skid_buffer`: - Decouples frontend and backend timing - Configurable depth per channel - 1-cycle latency overhead - Full backpressure handling

1.8.2 Pattern 2: Monitor Integration

Monitor modules combine functional core with verification: - Uses shared **axi_monitor_filtered** (rtl/amba/monitor/) - 2-level filtering (packet-type masks, then per-event-code masks; `err_select` is reserved and routes nothing) - 128-bit monitor bus + 64-bit side-band timestamp - Simplified for AXIL (MAX_TRANSACTION=8, vs 16 for the AXI4 wrappers)

1.8.3 Pattern 3: Clock Gating

Clock-gated variants (*_cg) add power management: - Dynamic gating based on channel valid activity plus the base module's busy - Activity is registered once (AXI4, AXI5, AXI4-Lite, AXI4-Stream) or twice (APB, APB5, AXI5-Stream) before reaching the ICG enable, which is combinational. AXI4 is single-stage, so the first usable gated-clock edge arrives 2 clocks after activity asserts — see the [Clock-Gated Variants Guide](#) - Better power savings than AXI4 (bursty register access)

1.9 Performance Characteristics

1.9.1 Throughput

Configuration	Rate	Notes
Single-beat read/write	1 txn/cycle	When buffers not stalled
Typical peripheral	1 txn/N cycles	N = backend response time

1.9.2 Latency

Latencies below are the module's own buffering overhead and **exclude** slave response time. Each channel traversed costs one clock in its skid buffer.

Path	Cycles	Notes
AR → R (single)	Slave latency + 2	1 clock in the AR buffer, 1 in the R buffer
AW+W → B (single)	Slave latency + 3	AW and W buffers, then the B buffer
Buffer overhead	+1 per channel	Skid buffer latency

The write figure assumes the slave accepts AW and W in the same cycle. If the slave's AWREADY and WREADY assert in different cycles, the write costs the later of the two handshakes instead.

1.9.3 Resource Usage

The figures below are **order-of-magnitude estimates only**. They are not measured synthesis results: no target device, tool version, or optimization setting is attached to them. Treat them as relative sizing between the variants, and run synthesis for your own device before budgeting area.

Module Type	LUTs	FFs	BRAM	Notes
Master/Slave (32-bit)	~200	~150	0	Per direction (rd/wr)
Monitor (+mon)	thousands	>5,000	0	At this family's default MAX_TRANSAC TIONS = 8. Structural floor counted from the RTL: transaction table 285 b x 8 = 2,280 FF, the reporter's local copy another 2,280, timeout timers 384, threshold latency 256
Clock-gated (+cg)	+50	+30	0	Clock gating logic

vs AXI4: roughly 40-50% smaller, on the same estimate basis (no ID tracking, no burst counters, simpler protocol). This is an estimate, not a measured comparison against the AXI4 modules.

1.10 Related Documentation

1.10.1 Protocol Specifications

- ARM IHI 0022E: AMBA AXI Protocol Specification
- Chapter 4: AXI4-Lite

1.10.2 RTL Design Sherpa Documentation

- [rtl-amba Overview](#) - Complete AMBA subsystem architecture
- [AXI4 Modules](#) - Full AXI4 protocol (comparison)
- [APB Modules](#) - Even simpler peripheral bus
- [GAXI Modules](#) - Generic AXI utilities (buffers, FIFOs)

1.10.3 Configuration and Integration

- [AXI Monitor Configuration Guide](#) - Monitor setup strategies
- [Monitor Packet Specification](#) - 128-bit packet plus 64-bit side-band timestamp

1.10.4 Source Code

- RTL: rtl/amba/axil4/
 - Monitor wrappers: rtl/amba/monitor/
 - Tests: val/amba/test_axil4*.py
 - Framework: bin/TBClasses/components/axil4/
 - Shared Infrastructure: rtl/amba/monitor/ (monitor base and filter),
rtl/amba/gaxi/ (gaxi_skid_buffer)
-

1.11 Design Notes

1.11.1 When to Use AXI4-Lite

✓ **Use AXI4-Lite For:** - Control and status registers (CSRs) - Peripheral configuration interfaces - Low-bandwidth control paths - Resource-constrained designs - Simpler protocol requirements

✗ **Use Full AXI4 For:** - High-bandwidth data transfers - Burst transactions required - Out-of-order support needed - DMA engines, memory controllers - Streaming data

1.11.2 Buffer Depth Guidelines

The `SKID_DEPTH_*` parameters are passed straight through to the `DEPTH` parameter of `gaxi_skid_buffer`, which is the **literal entry count**, not an exponent. `SKID_DEPTH_AR = 2` means a 2-entry buffer. `gaxi_skid_buffer` constrains `DEPTH` to one of 2..8 inclusive.

Address Channels (AR/AW): - Default: 2 entries - sufficient for most register access - Increase for high-latency address decode

Data Channels (R/W): - Default: 4 entries for R, 2 entries for W - R channel typically deeper (accommodate backend latency) - W channel can be shallower (single-beat)

Response Channel (B): - Default: 2 entries - responses are single-beat - Rarely needs adjustment

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2 AXIL4 Clock-Gated Variants Guide

Location: `rtl/amba/axil4/*_cg.sv` **Status:** ✓ Production Ready

2.1 Overview

All AXIL4 modules have clock-gated (`_cg`) variants that add power management through dynamic clock gating. Identical architecture to [AXI4 Clock Gating](#) but optimized for AXI4-Lite's simpler protocol.

2.1.1 Available Clock-Gated Modules

Module	Base Module	Description
<code>axil4_master_rd_cg</code>	axil4_master_rd	Clock-gated read master
<code>axil4_master_wr_cg</code>	axil4_master_wr	Clock-gated write master
<code>axil4_slave_rd_cg</code>	axil4_slave_rd	Clock-gated read slave

Module	Base Module	Description
axil4_slave_wr_cg	axil4_slave_wr	Clock-gated write slave
axil4_master_rd_mon_cg	axil4_master_rd_mon	Clock-gated read master + monitor
axil4_master_wr_mon_cg	axil4_master_wr_mon	Clock-gated write master + monitor
axil4_slave_rd_mon_cg	axil4_slave_rd_mon	Clock-gated read slave + monitor
axil4_slave_wr_mon_cg	axil4_slave_wr_mon	Clock-gated write slave + monitor

The four *_mon_cg modules live in rtl/amba/monitor/; the four plain *_cg modules live in rtl/amba/axil4/.

2.1.2 Key Features

- ✓ **Dynamic Clock Gating:** Automatic clock disable during idle
- ✓ **Configurable Idle Threshold:** Programmable idle count before gating
- ✓ **Full Functional Equivalence:** Identical to base modules
- ✓ **Status Monitoring:** cg_gating and cg_idle status outputs
- ✓ **Runtime Bypass:** cfg_cg_enable = 0 disables gating entirely

There is **no scan/test-mode bypass port**. clock_gate_ctrl exposes only cfg_cg_enable; a DFT flow must drive that low (or bypass the ICG cell in the scan-insertion step) to hold the clock free-running during scan.

2.2 Additional Parameters (All _cg Modules)

Parameter	Type	Default	Description
CG_IDLE_COUNT_WIDTH	int	4	Width of idle counter (max idle = 2 ^N -1 cycles)

All other parameters identical to base module.

2.3 Additional Ports (All _cg Modules)

2.3.1 Clock Gating Configuration

Port	Direction	Width	Description
cfg_cg_enable	Input	1	Enable clock gating (0=disabled, 1=enabled)
cfg_cg_idle_count	Input	CG_IDLE_COUNT_WIDTH	Idle cycles before gating

2.3.2 Clock Gating Status

Port	Direction	Width	Description
cg_gating	Output	1	Clock currently gated (1=gated, 0=running)
cg_idle	Output	1	No activity was seen on the previous cycle (registered ~wakeup)

All other ports are identical to the base module, with one exception: the base module's busy output is **not** brought out on the _cg wrapper. It is consumed internally as one of the wakeup terms. Use cg_idle (or cg_gating) for interface idle detection on a _cg module.

There is no cumulative gated-cycle counter on these modules. If you need one, count cg_gating in the surrounding logic on the ungated clock.

2.4 Usage Example

```
axil4_master_rd_cg #(
    // Base parameters
    .AXIL_ADDR_WIDTH(32),
    .AXIL_DATA_WIDTH(32),
    .SKID_DEPTH_AR(2),
    .SKID_DEPTH_R(4),

    // Clock gating
    .CG_IDLE_COUNT_WIDTH(4) // Up to 15 idle cycles
) u_axil_rd_cg (
    .aclk(axi_clk),
    .aresetn(axi_resetn),

    // Clock gating configuration
```

```

.cfg_cg_enable(1'b1),           // Enable gating
.cfg_cg_idle_count(4'd5),       // Gate after 5 idle cycles

// AXIL signals (identical to base module)
.fub_araddr(cpu_araddr),
// ... rest of AR/R signals ...

// Clock gating status
.cg_gating(rd_clk_gated),
.cg_idle(rd_idle)
);

```

2.5 Clock Gating Behavior

The `_cg` wrapper builds a wakeup term from the module's channel activity and the base module's internal busy, and hands it to `amba_clock_gate_ctrl`, which registers it and drives `clock_gate_ctrl` and its ICG cell. For `axil4_master_rd_cg` the terms are:

```

assign user_valid = fub_arvalid || fub_rready || int_busy;
assign axi_valid  = m_axil_arvalid || m_axil_rvalid;

```

Gating Conditions (All Must Be True): 1. No activity in `user_valid` or `axi_valid` 2. Idle for `cfg_cg_idle_count` consecutive cycles (the idle counter counts down to 0 and holds) 3. Gating enabled (`cfg_cg_enable = 1`)

Ungating Conditions (Any Triggers Ungating): 1. Any `*valid` signal asserted on either side of the module 2. `fub_rready` asserted, or the base module still busy

Gating latency: `cfg_cg_idle_count + 1` clocks after the internal wakeup deasserts, which is `cfg_cg_idle_count + 2` clocks after the last bus activity on AXI4-Lite (one extra for the `r_wakeup` flop).

Ungating latency: activity is registered once (AXI4, AXI5, AXI4-Lite, AXI4-Stream) or twice (APB, APB5, AXI5-Stream) before reaching the ICG enable, which is combinational. The `AXIL4_cg` wrappers drive `user_valid/axi_valid` combinational, so there is exactly **1 register stage** — `r_wakeup` inside `amba_clock_gate_ctrl` — and the first gated-clock rising edge available to the block arrives **2 clocks** after activity asserts, not 0. Note that `clock_gate_ctrl` contributes no flop of its own: `w_gate_enable` is combinational from wakeup straight into the ICG enable, so the second clock is the released edge itself rather than a further register stage. The wrapper covers this by forcing the module's ready outputs low while `cg_gating` is asserted, for example:

```

assign fub_arready  = cg_gating ? 1'b0 : int_arready;
assign m_axil_rready = cg_gating ? 1'b0 : int_rready;

```

so a request presented while gated is simply not accepted until the clock is running again. No data is lost, but a transaction that arrives into a gated interface pays the wakeup cost before its first handshake.

2.6 Configuration Guidelines

2.6.1 Idle Count Selection

Aggressive Gating (Register Access):

```
.cfg_cg_idle_count(4'd1)    // Gate after 1 idle cycle  
// AXI4-Lite typical: Burst register access with idle gaps
```

Moderate Gating (Balanced):

```
.cfg_cg_idle_count(4'd5)    // Gate after 5 idle cycles  
// General purpose configuration
```

Conservative Gating (Low Latency):

```
.cfg_cg_idle_count(4'd10)   // Gate after 10 idle cycles  
// Critical control paths, minimal overhead
```

Disable Gating:

```
.cfg_cg_enable(1'b0)        // Clock gating disabled  
// 100% utilization or ultra-low-latency paths
```

2.6.2 When to Use Clock Gating

✓ **Recommended For:** - Register access patterns (idle between accesses) - Low-duty-cycle control interfaces - Power-constrained systems

✗ **Not Recommended For:** - 100% utilization scenarios - Ultra-low-latency critical paths - Continuous register polling

2.7 Power Savings Estimates

2.7.1 Typical Savings (AXI4-Lite Register Access)

The percentages below are **rough expectations for dynamic clock power in the gated block only**, relative to the same module with `cfg_cg_enable = 0` at the same frequency. They are not measured power-analysis results, they exclude leakage (which gating does not reduce), and they exclude the clock tree upstream of the ICG cell. Run power analysis on your own target before relying on them.

Traffic Pattern	Duty Cycle	Idle Count	Est. Dynamic Power Savings
Burst register reads	20%	1	35-40%
Sporadic writes	10%	1	40-45%
Periodic polling	50%	5	20-25%
High activity	80%	5	5-10%

Note: AXI4-Lite often has better power savings than AXI4 due to: - Single-beat transactions (more idle gaps) - Simpler control (faster idle detection) - Register access patterns (bursty by nature)

2.8 Complete Documentation

For full clock gating details, see: - [AXI4 Clock Gating Guide](#) - Complete reference with: - Detailed gating behavior and state machine - Power savings calculations and examples - Dynamic configuration patterns - Synthesis considerations

AXIL4-specific notes: - Same architecture as AXI4 clock gating - Typically better power savings (bursty register access) - Simpler configuration (fewer outstanding transactions)

2.9 Related Documentation

2.9.1 Base Modules

- [axil4_master_rd](#) - Base read master
- [axil4_master_wr](#) - Base write master
- [axil4_slave_rd](#) - Base read slave
- [axil4_slave_wr](#) - Base write slave
- [Monitor modules](#) - Base monitor modules

2.9.2 Architecture

- [AXI4 Clock Gating Guide](#) - Complete reference
- [rtl-amba Overview](#) - Complete AMBA subsystem
- [AXIL4 Index](#) - AXIL4 module index

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3 AXIL4 Master Read

Module: axil4_master_rd.sv **Location:** rtl/amba/axil4/ **Status:** ✓ Production Ready

3.1 Overview

The AXIL4 Master Read module provides a buffered AXI4-Lite read interface for master devices. AXI4-Lite is a simplified subset of AXI4 designed for control register access with reduced signal count and single-beat transactions only.

3.1.1 Key Features

- ✓ **AXI4-Lite Protocol:** Simplified read-only interface (AR and R channels)
 - ✓ **Single-Beat Transactions:** No burst support (always 1 transfer)
 - ✓ **Buffered Channels:** Configurable skid buffers for timing closure
 - ✓ **Elastic Buffering:** Decouples frontend and backend timing
 - ✓ **Activity Monitoring:** Busy signal for clock gating
 - ✓ **Minimal Latency:** 1-cycle buffer overhead per channel
-

3.2 Parameters

Parameter	Type	Default	Description
AXIL_ADDR_WIDTH	int	32	Address bus width (typically 32)
AXIL_DATA_WIDTH	int	32	Data bus width (32 or 64)
SKID_DEPTH_AR	int	2	AR channel skid buffer

Parameter	Type	Default	Description
SKID_DEPTH_R	int	4	depth, in entries R channel skid buffer depth, in entries

3.2.1 Derived Parameters

Each channel's skid buffer stores the whole channel payload as one packed vector, so the derived *Size parameters are just the sum of the widths of the signals concatenated into that buffer.

Parameter	Calculation	Description
AW	AXIL_ADDR_WIDTH	Internal address width alias
DW	AXIL_DATA_WIDTH	Internal data width alias
ARSize	AW+3	AR packet size: {araddr[AW-1:0], arprot[2:0]} — the 3 is ARPROT
RSize	DW+2	R packet size: {rdata[DW-1:0], rresp[1:0]} — the 2 is RRESP

3.3 Port Groups

3.3.1 Clock and Reset

Port	Direction	Width	Description
aclk	Input	1	AXI clock
aresetn	Input	1	Active-low asynchronous reset

3.3.2 Frontend AXI4-Lite Read Interface (Input Side)

Read Address Channel (AR): | Port | Direction | Width | Description | |——|———|———|
 ————| | fub_araddr | Input | AW | Read address | | fub_arprot | Input | 3 | Protection

attributes | | fub_arvalid | Input | 1 | Address valid | | fub_arready | Output | 1 | Address ready (from buffer) |

Read Data Channel (R): | Port | Direction | Width | Description | |——|———|——|
 ————| | fub_rdata | Output | DW | Read data | | fub_rresp | Output | 2 | Read response (OKAY, SLVERR, DECERR) | | fub_rvalid | Output | 1 | Data valid (from buffer) | | fub_rready | Input | 1 | Data ready |

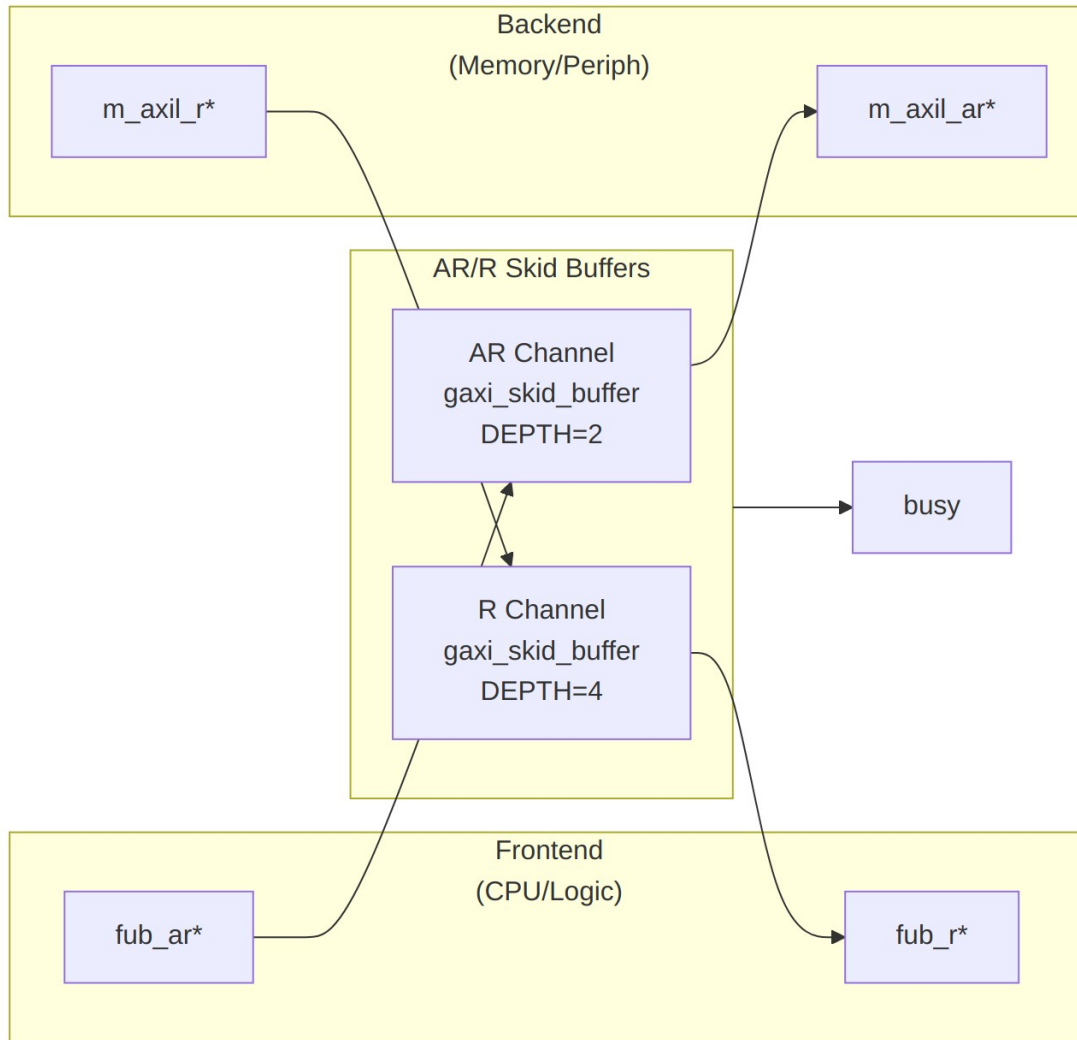
3.3.3 Backend AXI4-Lite Master Interface (Output Side)

Read Address Channel (AR): | Port | Direction | Width | Description | |——|———|——|
 ————| | m_axil_araddr | Output | AW | Read address (to slave) | | m_axil_arprot | Output | 3 | Protection attributes | | m_axil_arvalid | Output | 1 | Address valid (from buffer) | | m_axil_arready | Input | 1 | Address ready (from slave) |

Read Data Channel (R): | Port | Direction | Width | Description | |——|———|——|
 ————| | m_axil_rdata | Input | DW | Read data (from slave) | | m_axil_rresp | Input | 2 | Read response | | m_axil_rvalid | Input | 1 | Data valid (from slave) | | m_axil_rready | Output | 1 | Data ready (to slave) |

3.3.4 Status Outputs

Port	Direction	Width	Description
busy	Output	1	Interface active (for clock gating)



Module Architecture

3.4 Signal Flow

3.4.1 Read Transaction Sequence

Cycle 1: `fub_araddr=0x1000`, `fub_arvalid=1`, `fub_arprot=0`
`fub_arready=1` (buffer accepts)

Cycle 2: AR buffer forwards to backend:
`m_axil_araddr=0x1000`, `m_axil_arvalid=1`
`m_axil_arready=1` (slave accepts)

Cycle 3-5: Slave processes read

Cycle 6: m_axil_rdata=0xDEADBEEF, m_axil_rresp=OKAY
m_axil_rvalid=1, m_axil_rready=1 (R buffer accepts)

Cycle 7: R buffer forwards to frontend:
fub_rdata=0xDEADBEEF, fub_rresp=OKAY
fub_rvalid=1, fub_rready=1 (frontend accepts)

3.5 Usage Example

3.5.1 Basic Integration

```
axil4_master_rd #(
    .AXIL_ADDR_WIDTH(32),
    .AXIL_DATA_WIDTH(32),
    .SKID_DEPTH_AR(2),      // 2 entries
    .SKID_DEPTH_R(4)        // 4 entries
) u_axil_master_rd (
    .aclk          (axi_clk),
    .aresetn       (axi_resetn),

    // Frontend interface (from CPU)
    .fub_araddr    (cpu_araddr),
    .fub_arprot    (cpu_arprot),
    .fub_arvalid   (cpu_arvalid),
    .fub_arready   (cpu_arready),

    .fub_rdata     (cpu_rdata),
    .fub_rresp     (cpu_rresp),
    .fub_rvalid    (cpu_rvalid),
    .fub_rready    (cpu_rready),

    // Backend interface (to peripheral/memory)
    .m_axil_araddr (periph_araddr),
    .m_axil_arprot (periph_arprot),
    .m_axil_arvalid(periph_arvalid),
    .m_axil_arready(periph_arready),

    .m_axil_rdata  (periph_rdata),
    .m_axil_rresp  (periph_rresp),
    .m_axil_rvalid (periph_rvalid),
    .m_axil_rready (periph_rready),

    // Status
```

```
        .busy                (rd_busy)
    );
```

3.5.2 Register Read Example (Testbench Stimulus)

The snippet below is **testbench stimulus, not RTL**. It uses an `initial` block and blocking assignments to drive the frontend ports and is not synthesizable. In a real design the frontend handshake is driven by an `always_ff` block or by the register-access state machine of the requesting logic.

```
// Read from control register at 0x1000
initial begin
    @(posedge aclk);
    fub_araddr = 32'h1000;
    fub_arprot = 3'b000; // Unprivileged, secure, data access
    fub_arvalid = 1'b1;

    @(posedge aclk);
    while (!fub_arready) @(posedge aclk); // Wait for buffer accept
    fub_arvalid = 1'b0;

    // Wait for read data
    fub_rready = 1'b1;
    @(posedge aclk);
    while (!fub_rvalid) @(posedge aclk);

    $display("Read data: 0x%08x, RRESP: %0d", fub_rdata, fub_rresp);
end
```

3.6 Design Notes

3.6.1 AXI4-Lite Protocol Constraints

Single-Beat Only: - No burst support (implicitly `ARLEN=0`, `ARSIZE=log2(DW/8)`) - Each AR transaction results in exactly 1 R transaction - No `RLAST` signal (always last)

No Out-of-Order: - No `ARID`/`RID` signals - Responses always in order - Simpler transaction tracking

Reduced Signals: - No `ARCACHE`, `ARQOS`, `ARREGION`, `ARLOCK`, `ARSIZE`, `ARLEN`, `ARBURST` - Only `ARADDR` and `ARPROT` for control

3.6.2 Buffer Depth Selection

SKID_DEPTH_AR and SKID_DEPTH_R are passed straight through to the DEPTH parameter of gaxi_skid_buffer. DEPTH is the **literal entry count**, not an exponent, and is constrained to one of 2..8 inclusive.

AR Channel (Addresses): - SKID_DEPTH_AR = 2 (2 entries): Typical for control registers - Increase if many back-to-back reads with slow slave response

R Channel (Data): - SKID_DEPTH_R = 4 (4 entries): Accommodates pipeline latency - Increase for high-latency backends (DRAM, off-chip peripherals)

3.6.3 Busy Signal

busy is the OR of two persistent terms and two transient terms:

Term	Kind	Condition
AR buffer occupancy	Persistent	int_ar_count > 0
R buffer occupancy	Persistent	int_r_count > 0
Frontend presenting a new address	Transient	fub_arvalid
Backend presenting new data	Transient	m_axil_rvalid

The two transient terms can be high for a single cycle, so busy itself can pulse. This is intentional: the clock-gate controller does not gate on busy directly. amba_clock_gate_ctrl registers activity into a wakeup flop and then requires cfg_cg_idle_count consecutive idle cycles before gating, so a one-cycle handshake reloads the idle counter rather than causing the gate to oscillate. Consumers of busy outside the clock-gating path should qualify or stretch it themselves if they need a level signal.

Use for: - Clock gating control (see axil4_master_rd_cg) - Power management - Interface idle detection

3.6.4 PROT Encoding

ARPROT[2:0]	Privilege	Security	Type
3'b000	Unprivileged	Secure	Data
3'b001	Unprivileged	Secure	Instruction
3'b010	Unprivileged	Non-secure	Data
3'b011	Unprivileged	Non-secure	Instruction
3'b100	Privileged	Secure	Data
3'b101	Privileged	Secure	Instruction

ARPROT[2:0]	Privilege	Security	Type
3'b110	Privileged	Non-secure	Data
3'b111	Privileged	Non-secure	Instruction

Typical: 3'b000 (unprivileged secure data) for most peripherals

3.7 Performance Characteristics

3.7.1 Latency

Path	Cycles	Notes
Frontend → Backend (AR)	1	Skid buffer overhead
Backend → Frontend (R)	1	Skid buffer overhead
Total read latency	Slave latency + 2	AR + Slave + R

3.7.2 Throughput

Scenario	Rate	Notes
Back-to-back reads	1 read/cycle	If buffers not stalled
Typical peripheral	1 read/N cycles	N = slave response time

3.7.3 Resource Usage

These are **order-of-magnitude estimates**, not measured synthesis results — no target device, tool version, or optimization setting is attached to them. They scale with AXIL_DATA_WIDTH and the SKID_DEPTH_* settings. Synthesize for your own device before budgeting area.

Resource	Count	Notes
LUTs	~200	Estimate, 32-bit data, default depths
FFs	~150	Buffer state + control
BRAM	0	Skid buffers are flop-based

3.8 Related Modules

3.8.1 Core Modules

- [axil4_master_wr](#) - Write master counterpart
- [axil4_slave_rd](#) - Slave read interface
- [axil4_slave_wr](#) - Slave write interface

3.8.2 Monitor Modules

- [axil4_master_rd_mon](#) - Master read with monitoring (rtl/amba/monitor/)

3.8.3 Clock-Gated Variants

- [axil4_master_rd_cg](#) - Clock-gated version

3.8.4 Supporting Infrastructure

- [gaxi_skid_buffer](#) - Elastic buffering (rtl/amba/gaxi/)
-

3.9 References

3.9.1 Specifications

- ARM IHI 0022E: AMBA AXI and ACE Protocol Specification
- Chapter 4: AXI4-Lite (simplified subset)

3.9.2 Documentation

- [rtl-amba Overview](#) - Complete AMBA subsystem
- [AXI4 Master Read](#) - Full AXI4 comparison

3.9.3 Source Code

- RTL: rtl/amba/axil4/axil4_master_rd.sv
 - Tests: val/amba/test_axil4_master_rd.py
 - Framework: bin/TBClasses/components/axil4/
-

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4 AXIL4 Master Read Interface (Clock-Gated)

Module: axil4_master_rd_cg.sv **Base Module:** [axil4_master_rd](#) **Location:** rtl/amba/axil4/ **Status:** ✓ Production Ready

4.1 Quick Reference

This is the **clock-gated variant** of [axil4_master_rd](#).

For complete clock-gating documentation, usage examples, and configuration guidelines, see:

- [AXIL4 Clock-Gated Variants Guide](#) (AXIL4-specific)
 - [Clock-Gated Variants Guide](#) (cross-protocol overview)
-

4.2 Summary

The axil4_master_rd_cg module adds power optimization to axil4_master_rd through activity-based clock gating:

- ✓ **Same Functionality:** 100% equivalent to base module
 - ✓ **Power Savings:** dynamic clock power in the gated block scales with idle fraction; see the [AXIL4 Clock-Gated Variants Guide](#) for the estimate table and its caveats
 - ✓ **Configurable:** idle threshold and enable, both at **runtime** via `cfg_cg_idle_count / cfg_cg_enable`
 - ✓ **Zero Overhead When Disabled:** `cfg_cg_enable = 0` holds the clock free-running, making behavior identical to the base module
-

4.3 Common Parameters

In addition to all `axil4_master_rd` parameters:

The `_cg` wrapper adds exactly **one** parameter. Gating is enabled and tuned at runtime through ports, not through parameters — there is no `ENABLE_CLOCK_GATING` parameter, no `CG_IDLE_CYCLES` parameter, and no per-domain `CG_GATE_*` parameters on this module.

Parameter	Type	Default	Description
<code>CG_IDLE_COUNT_WIDTH</code>	<code>int</code>	<code>4</code>	Width of the idle counter; max idle count = $2^N - 1$

4.3.1 Additional Ports

Port	Direction	Width	Description
<code>cfg_cg_enable</code>	Input	<code>1</code>	Enable clock gating (0 = clock always running)
<code>cfg_cg_idle_count</code>	Input	<code>CG_IDLE_COUNT_WIDTH</code>	Idle cycles before gating
<code>cg_gating</code>	Output	<code>1</code>	Clock currently gated
<code>cg_idle</code>	Output	<code>1</code>	No activity on the previous cycle

The base module's busy output is **not** exposed on the `_cg` wrapper; it is consumed internally as a wakeup term. All other ports are identical to `axil4_master_rd`.

4.4 Quick Usage

```
axil4_master_rd_cg #(
    // Base module parameters (see axil4_master_rd.md)
    .AXIL_ADDR_WIDTH(32),
    .AXIL_DATA_WIDTH(32),
    .SKID_DEPTH_AR(2),
    .SKID_DEPTH_R(4),

    // Clock gating (see CG guide for details)
    .CG_IDLE_COUNT_WIDTH(4)
) u_cg (
    .aclk(clk),
    .aresetn(rst_n),
```

```
// Clock gating control (runtime, not parameters)
.cfg_cg_enable(1'b1),
.cfg_cg_idle_count(4'd5),

// ... all other ports same as axil4_master_rd, except `busy`,
// which is replaced by cg_gating / cg_idle
.cfg_gating(clk_gated),
.cfg_idle(if_idle)
);
```

4.5 Documentation

- **Base Module Functionality:** [axil4_master_rd.md](#)
 - **AXIL4 Clock Gating Guide:** [axil4_clock_gating_guide.md](#)
 - **Cross-Protocol Clock Gating Guide:** [clock_gated_variants.md](#)
 - **Detailed CG Examples:**
 - [axi4_master_rd_mon_cg.md](#) (AXI4 monitor)
 - [axil4_master_rd_mon_cg.md](#) (AXIL4 monitor)
 - [apb4_slave_cg.md](#) (APB interface)
-

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5 AXIL4 Master Read with Monitoring

Module: `axil4_master_rd_mon.sv` **Location:** `rtl/amba/monitor/` **Status:** ✓ Production Ready

5.1 Overview

Combines [axil4_master_rd](#) with the core **axi_monitor_filtered** for transaction monitoring. Simplified for AXI4-Lite (single-beat, no burst, fixed ID=0).

5.1.1 Key Features

- ✓ All features of base **axil4_master_rd** module
- ✓ **Integrated Monitoring:** Uses shared **axi_monitor_filtered** (rtl/amba/monitor/)
- **2-Level Filtering:** packet-type masks, then per-event-code masks. **err_select** is RESERVED – it feeds only the conflict check and routes nothing.
- ✓ **Error Detection:** Protocol violations, timeouts, orphans
- ✓ **128-bit Monitor Bus:** Standardized packet format paired with 64-bit side-band timestamp
- ✓ **Reduced Complexity:** MAX_TRANSACTIONS=8 (vs 16-32 for AXI4)

5.2 Additional Parameters (Beyond Base Module)

Parameter	Type	Default	Description
UNIT_ID	logic [7:0]	8'h01	8-bit unit identifier emitted in the <code>unit_id</code> packet field
AGENT_ID	logic [15:0]	16'h000A	16-bit agent identifier emitted in the <code>agent_id</code> packet field
MAX_TRANSACTIONS	int	8	Max outstanding transactions. Reduced for AXI4-Lite; the AXI4 wrappers default to 16.
ACTIVE_TRANS_THRES HOLD	int	MAX_TRANSACTIONS/2	Active-transaction count that trips a threshold packet when <code>cfg_threshold_enable=1</code> . Replaces the former hardwired 8/4; threshold packets now scale with

Parameter	Type	Default	Description
ENABLE_FILTERING	bit	1	the table sizing Enable packet filtering: two active drop levels (packet type, then event code). Level 2 is reserved and routes nothing
ADD_PIPELINE_STAGE	bit	0	Add register stage for timing closure
USE_MONITOR	bit	1	Synthesis-time monitor enable. 0 = omit monitor and tie outputs to safe non-blocking defaults; 1 = full monitor functionality.
N_ADDR_RANGES	int	0	Number of address-range comparators. 0 = checker omitted (zero area). >0 = N independent [low, high] ranges; feeds the shared allowlist checker: a debug-range hit -> AddrMatch, an error-allowlist miss -> Error/ADDR_RANGE (see axi_monitor_addr_check.md).

ACLK_MHZ | int | 100 | Clock frequency in MHz. Builds the microsecond tick LUT in counter_freq_invariant. **Leave this at 100 on a 90 MHz part and every us-denominated timeout is wrong, silently.** |

CFI_MIN_FREQ_MHZ / CFI_MAX_FREQ_MHZ | int | ACLK_MHZ | Bounds of the freq-invariant counter LUT (cfg_freq_sel indexes within them). Equal bounds means one entry and cfg_freq_sel has no effect. |

USE_WDATA_ORDER_Q | bit | 0 | Write-data ordering queue, forwarded to axi_monitor_base. |

NUM_BANKS | int | 1 | Banked transaction tables. **>1 on a WRITE monitor requires**

USE_WDATA_ORDER_Q=1 – axi_monitor_trans_mgr fails elaboration otherwise. |

5.2.1 Synthesis-Cone Parameters

Each detection cone can be compiled out to save area. These are forwarded to `axi_monitor_base`; by default the classic cones are on and the debug cone is off.

Parameter	Type	Default	Effect when 0
ENABLE_ERROR_LOGIC	bit	1	Drop the error-detection cone
ENABLE_TIMEOUT_LOGIC	bit	1	Drop the timeout cone and the <code>axi_monitor_timeout</code> instance
ENABLE_COMPL_LOGIC	bit	1	Drop the completion cone
ENABLE_THRESHOLD_LOGIC	bit	1	Drop the threshold cone
ENABLE_PERF_LOGIC	bit	1	Gates only the reporter's legacy perf-packet cone and the two lifetime counters – the window FSM and meters are unconditional
ENABLE_DEBUG_LOGIC	bit	0	(off by default) drop the debug cone

The former `CAM_PIPELINE` / `TRANS_CAM_PIPELINE` parameters were removed — the transaction CAM is now always pipelined. Inside this wrapper `ENABLE_PERF_PACKETS` is tied to `1'b1` (perf packet path always instantiated, gated by `ENABLE_PERF_LOGIC`) and `ENABLE_DEBUG_MODULE` is tied to `1'b0`.

5.3 Performance Monitoring

When performance monitoring is enabled, the wrapper forwards a **measurement-window state machine** plus a bank of R-channel (read-data) utilization counters to `axi_monitor_base`. All counters accumulate **only while a window is open** (`window_active = 1`) and hold their values between windows so the host can read a completed window's totals. The counters advance only while `cfg_perf_enable = 1`. `ENABLE_PERF_LOGIC = 0` does NOT drop them: the window FSM and its counters are unconditional always_ff blocks in `axi_monitor_base`, outside every generate. That parameter gates only `g_perf` in the reporter – the legacy perf-packet cone and the two lifetime counters. `USE_MONITOR = 0` is what ties the perfmon outputs off.

Avoid enabling completion (`cfg_compl_enable`) and performance (`cfg_perf_enable`) packets simultaneously under heavy traffic — the monitor bus sustains at most one packet per two cycles. Runtime-disabling either class is safe (terminal entries auto-retire; see [axi_monitor_reporter](#)); alternatively, `cfg_axi_pkt_mask` drops the packets while keeping marking and counting. See `docs/user-guides/AXI_Monitor_Configuration_Guide.md`.

5.3.1 The Measurement Window

A window is opened by a **start event** and closed by an **end event**:

- `cfg_start_event_sel` / `cfg_end_event_sel` (3-bit) select the event source (e.g. `3'b010` selects the `cfg_perf_enable` edge).
- `cfg_start_trigger` / `cfg_end_trigger` pulses fire the window directly from an engine or CSR.
- `cfg_window_force_close` is a software override that closes the window immediately.

While the window is open, `window_active` is high and `window_cycles` [31:0] free-runs, counting every clock elapsed inside the window.

5.3.2 Utilization Counters (R channel)

Every in-window cycle is classified by the R-channel valid/ready into exactly one of four buckets:

Output	Width	Condition	Meaning
<code>perf_prod_cycles</code>	32	<code>rvalid && rready</code>	productive beat transferred
<code>perf_bp_cycles</code>	32	<code>rvalid && !rready</code>	back-pressure (data offered, consumer not ready)
<code>perf_starv_cycles</code>	32	<code>!rvalid && rready</code>	starvation (consumer ready, no valid data)
<code>perf_idle_cycles</code>	32	<code>!rvalid && !rready</code>	idle

The four buckets sum to `window_cycles`, so `utilization = perf_prod_cycles / window_cycles`.

5.3.3 Throughput Counters

Output	Width	Meaning
perf_beat_count	32	data beats transferred (= perf_prod_cycles, 1 beat/cycle)
perf_byte_count	64	bytes transferred = beats × (1 << AxSIZE); AXIL is fixed at AxSIZE=3 'b010 (4 bytes)
perf_burst_count	32	AR (read-address) handshakes

AXI4-Lite note: every transaction is a single data beat (ARLEN is implicitly 0), so perf_burst_count counts AR handshakes = transactions and perf_beat_count equals the transaction count. Average burst length (perf_beat_count / perf_burst_count) is therefore always 1.

5.4 Monitor Backpressure (block_ready)

block_ready is an internal flow-control net inside the wrapper – it is not a port. It goes low when the monitor’s transaction-table occupancy reaches its blocking threshold (a function of MAX_TRANSACTIONS; the reporter FIFO depth INTR_FIFO_DEPTH has no path to it). The wrapper ANDs it into the upstream-facing fub_axil_arready so a saturated monitor throttles new transactions at the handshake instead of dropping events.

- **Where the stall lands:** the upstream fub_axil_arready is forced low until the monitor drains.
- **When USE_MONITOR=0:** block_ready is internally tied high, so the wrapper imposes no stall and runs at full bandwidth.
- **When cfg_monitor_enable=0:** the wrapper gate forces the upstream ready open, so a runtime-disabled monitor can never stall the datapath (the AXI4 monitors assert this as an in-RTL formal property, ap_disabled_never_stalls; this module has no ifdef FORMAL block of its own, so here the guarantee rests on the gate expression above, not a proof).

Recovery is guaranteed by the **saturation-recovery contract**: command-originated table entries are capped at MAX_TRANSACTIONS - cmd_entry_reserve(MAX_TRANSACTIONS) (reserve = 4 for tables of 16 or more, 0 below; the function lives in monitor_common_pkg), and block_ready re-asserts at occupancy < MAX_TRANSACTIONS - (reserve - 1) – a threshold strictly ABOVE the command cap – so a saturated table always drains back below the reopen point. Blocking

throttles; it never deadlocks. Tables smaller than 16 keep full legacy allocation (small tables cannot spare slots) and trade the recovery guarantee for tracking capacity. The contract is verified by in-RTL formal properties (mutation-checked) and a 100-seed deliberately-undersized-table stream sweep; see [axi_monitor_base](#) for the canonical description.

Sizing note: a monitor on a bus shared by several channels/requesters must size MAX_TRANSACTIONS to cover NUM_CHANNELS x per-channel outstanding plus margin – the per-channel limit alone makes the monitor throttle the shared master. Tables deeper than 64 also need Verilator's `--unroll-count` raised (default 64) in sim builds.

5.5 Address-Range Checker

With `N_ADDR_RANGES > 0` the wrapper instantiates the shared allowlist checker (`axi_monitor_addr_check`). Each range carries a DEBUG/ERROR flavor:

- **DEBUG range** — a hit emits a `PktTypeAddrMatch` (4'h8) packet with event code `AXI_ADDR_RANGE_MATCH` (8'h01), gated by `cfg_debug_enable`.
- **ERROR range** — the enabled ERROR ranges form an allowlist; an address in NONE of them emits a `PktTypeError` (4'h0) packet with event code `AXI_ERR_ADDR_RANGE` (8'h0D), gated by `cfg_error_enable`.

This wrapper does not expose `ADDR_RANGE_IS_ERROR`; all ranges default to the DEBUG (match) flavor.

Config inputs (active only when `N_ADDR_RANGES > 0`): - `cfg_addr_check_enable` — master on/off for the checker. - `cfg_addr_range_enable[N-1:0]` — per-range enable bit. - `cfg_addr_range_low/high[N-1:0][AXI_ADDR_WIDTH-1:0]` — inclusive bounds.

Event encoding: `event_data[63:60]` = `range_index` (matching DEBUG range, or 4'hF sentinel on an ERROR miss); `event_data[59:0]` = full address. **Filtering:** `AddrMatch` is dropped by `cfg_axi_addr_mask[1]`, the `ADDR_RANGE` error by `cfg_axi_error_mask[13]`. See the checker page for coalescing + formal properties.

5.5.1 Packet filters

Two independent runtime filters cut monbus traffic without touching the datapath. Both are inert until driven, which is the failure to watch for: the build-time parameter only decides whether the logic is SYNTHESISED, and a design that sets the parameter but leaves the `cfg_*` ports tied low filters nothing and looks like the feature is broken.

Address-range filter — `ADDR_FILTER_ENABLE` (bit, default 0) builds it.

Port	Width	Description
cfg_addr_filter_enable	1	High: suppress packets for transactions outside the window. Low: inert, whatever the parameter says
cfg_addr_filter_low	ADDR_WIDTH	Window base, inclusive
cfg_addr_filter_high	ADDR_WIDTH	Window limit, inclusive

The verdict is latched per table entry at ALLOCATION, from the command address, and held for that entry's life. Widening the window mid-flight does not un-filter entries already admitted — which is what makes it safe to reprogram live. Contrast the address-range CHECKER above, which re-evaluates per command.

There is no runtime ID filter here. AXI-Lite has no transaction IDs, so this wrapper ties `cfg_id_filter_enable` / `cfg_id_match_base` / `cfg_id_match_count` off on the inner monitor rather than exposing them — a filter keyed on a field the protocol lacks has nothing to match against.

5.6 Additional Ports (Beyond Base Module)

5.6.1 Monitor Configuration

Port	Direction	Width	Description
cfg_monitor_enable	Input	1	Master runtime gate. 0 = monitor inert: no allocation, transaction CAM held clear, and the upstream ready is never stalled (a disabled monitor cannot block the datapath). 1 = normal operation
cam_clear	Input	1	Synchronous clear of the monitor transaction CAM (driven from the harness clear control bit, e.g. CTRL[4])
cfg_error_enable	Input	1	Enable error packets

Port	Direction	Width	Description
cfg_timeout_enable	Input	1	Enable timeout detection
cfg_completion_enable	Input	1	Enable transaction-completion packets
cfg_threshold_enable	Input	1	Enable threshold-crossed packets
cfg_performance_enable	Input	1	Enable performance packets
cfg_debug_enable	Input	1	Enable debug/trace packets (the 6th “debug cone” reporter sub-block)
cfg_timeout_cycles	Input	16	Unified coarse timeout control: 0 = legacy full-scale (15 ticks), 1-15 = that many timer ticks per phase, >15 saturates at 15. One value drives all three phase counts (addr/data/resp), measured in cfg_freq_sel-scaled timer ticks, not raw clock cycles
cfg_latency_threshold	Input	32	Latency alert threshold

The debug cone is the 6th reporter sub-block (axi_monitor_reporter_debug), enabled by cfg_debug_enable and synthesized only when ENABLE_DEBUG_LOGIC = 1. The related cfg_debug_level (4b) and cfg_debug_mask (16b) inputs of axi_monitor_base are **not** exposed on this AXIL wrapper — they are tied to 4'h0 / 16'h0 internally; cfg_active_trans_threshold is driven from the ACTIVE_TRANS_THRESHOLD parameter (default MAX_TRANSACTIONS/2).

5.6.2 Performance Monitoring Ports

Port	Direction	Width	Description
cfg_start_event_sel	Input	3	Window start event source select
cfg_end_event_sel	Input	3	Window end event source select
cfg_start_trigger	Input	1	Pulse: open the measurement window
cfg_end_trigger	Input	1	Pulse: close the measurement window
cfg_window_force_close	Input	1	Software override: force the window closed
window_active	Output	1	High while a measurement window is open
window_cycles	Output	32	Cycles elapsed in the current window
perf_prod_cycles	Output	32	valid && ready cycles
perf_bp_cycles	Output	32	valid && !ready cycles (back-pressure)
perf_starv_cycles	Output	32	!valid && ready cycles (starvation)
perf_idle_cycles	Output	32	!valid && !ready cycles
perf_beat_count	Output	32	data beats transferred
perf_byte_count	Output	64	bytes transferred
perf_burst_count	Output	32	AR (address-phase) handshakes

When perfmon is unused, the integrating block ties cfg_start_event_sel/cfg_end_event_sel to 3'b111 and the remaining cfg_* perfmon inputs to 0. When USE_MONITOR = 0 all perfmon outputs are tied to 0.

5.6.3 Filtering Masks (9 masks total)

All nine are 16-bit inputs. `cfg_axi_pkt_mask` is indexed by the 4-bit `packet_type` field; the per-type event masks are indexed by the low nibble of `event_code`. In every case a **set bit drops** the matching packet.

Port	Width	Description
<code>cfg_axi_pkt_mask</code>	16	Level 1: drop mask indexed by <code>packet_type</code>
<code>cfg_axi_err_select</code>	16	Route selected packet types to the error path. Overlapping a bit with <code>cfg_axi_pkt_mask</code> raises <code>cfg_conflict_error</code> .
<code>cfg_axi_error_mask</code>	16	Level 3: mask individual error events
<code>cfg_axi_timeout_mask</code>	16	Level 3: mask individual timeout events
<code>cfg_axi_completion_mask</code>	16	Level 3: mask individual completion events
<code>cfg_axi_threshold_mask</code>	16	Level 3: mask individual threshold events
<code>cfg_axi_performance_mask</code>	16	Level 3: mask individual performance events
<code>cfg_axi_address_match_mask</code>	16	Level 3: mask individual address-match events
<code>cfg_axi_debug_mask</code>	16	Level 3: mask individual debug events

There is no `cfg_axi_full_mask` port.

5.6.4 Monitor Bus Output

Port	Direction	Width	Description
<code>monbus_valid</code>	Output	1	Monitor packet valid
<code>monbus_ready</code>	Input	1	Downstream ready
<code>monbus_packet</code>	Output	128	<code>monitor_packet_t</code> (128-bit format)
<code>monbus_timestamp</code>	Output	64	<code>monbus_timestamp_t</code>

Port	Direction	Width	Description
mestamp			paired atomically with monbus_packet
i_mon_time	Input	64	Free-running counter from monbus_group_core, sampled at packet emission

5.6.5 Status

Port	Direction	Width	Description
busy	Output	1	Interface active
active_transactions	Output	8	Current outstanding count
error_count	Output	16	Lifetime count of error+timeout packets actually emitted (reporter perf counter; reads 0 when ENABLE_PERF_LOGIC=0 or USE_MONITOR=0)
transaction_count	Output	32	Lifetime count of completion packets actually emitted (zero-extended 16-bit reporter perf counter; reads 0 when ENABLE_PERF_LOGIC=0 or USE_MONITOR=0)
cfg_conflict_error	Output	1	Set when cfg_axi_pkt_mask and cfg_axi_err_select overlap

5.7 Usage Example

```
axil4_master_rd_mon #(
    // Base parameters
    .AXIL_ADDR_WIDTH(32),
    .AXIL_DATA_WIDTH(32),
    .SKID_DEPTH_AR(2),
    .SKID_DEPTH_R(4),

    // Monitor parameters
    .UNIT_ID(1),
    .AGENT_ID(10),
    .MAX_TRANSACTIONS(8),
    .ENABLE_FILTERING(1)
) u_axil_rd_mon (
    .aclk(axi_clk),
    .aresetn(axi_resetn),

    // AXIL interfaces (same as axil4_master_rd)
    .fub_axil_araddr(cpu_araddr),
    // ... other AR/R signals ...

    // Monitor configuration
    .cfg_monitor_enable(1'b1),
    .cfg_error_enable(1'b1),
    .cfg_timeout_enable(1'b1),
    .cfg_perf_enable(1'b0),           // Avoid congestion
    .cfg_timeout_cycles(16'd10),     // 10 timer ticks per phase (>15
saturates)
    .cfg_latency_threshold(32'd500),

    // Filtering masks - a SET bit DROPS that packet type
    .cfg_axi_pkt_mask(16'b1111_1111_1111_0110), // Keep Error (bit 0)
+ Timeout (bit 3)
    // ... other masks ...

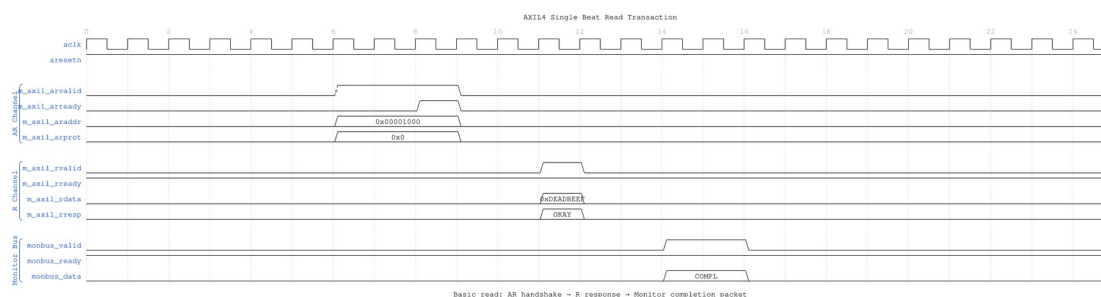
    // Monitor bus
    .monbus_valid(mon_valid),
    .monbus_ready(mon_ready),
    .monbus_packet(mon_data),
    .monbus_timestamp(mon_time)
);
```

Mask polarity. `cfg_axi_pkt_mask` is 16 bits because it is indexed by the 4-bit `packet_type` field — one bit per packet type — not by the 128-bit packet width. `axi_monitor_filtered` computes `pkt_drop = cfg_axi_pkt_mask[pkt_type]`, so a set bit **drops** that type. To pass only `PktTypeError` (4'h0) and `PktTypeTimeout` (4'h3), clear bits 0 and 3 and set the rest:

16'b1111_1111_1111_0110. See the [Monitor Packet Specification](#) for the full packet-type enum.

5.8 Timing Diagrams

5.8.1 Scenario 1: Single-Beat Read Transaction

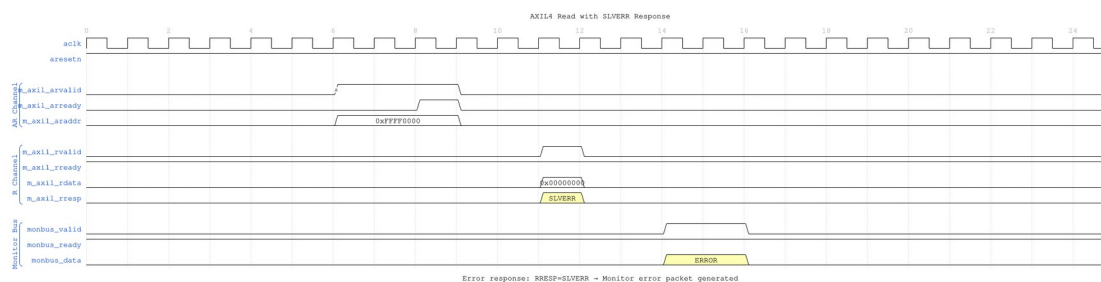


Single Beat Read

WaveJSON: [single_beat_read_001.json](#)

Key Observations: - AR channel handshake: ARVALID asserted, ARREADY responds - R channel response: Slave returns data with RRESP=OKAY - Monitor generates completion packet when R phase completes - Single-beat transaction: No burst length (implicit ARLEN=0)

5.8.2 Scenario 2: Read Error (SLVERR)



Read Error SLVERR

WaveJSON: [read_error_slvrr_001.json](#)

Key Observations: - Invalid address triggers RRESP=SLVERR - Monitor detects error response and generates ERROR packet - Transaction completes despite error (data may be undefined) - Error packet includes address and response code

[illegible]

vs Full AXI4 Monitoring: - **Fixed ID:** Always ID=0 (no out-of-order tracking) - **Single-Beat:** No burst handling (ARLEN implicitly 0) - **Reduced Tables:** MAX_TRANSACTION=8 (vs 16-32 for AXI4) - **Same Infrastructure:** Uses axi_monitor_filtered like AXI4

5.10.1 Base Module

- [axil4_master_rd](#) - Functional module documentation

5.10.2 Monitor Infrastructure

- [AXI4 Master Read Mon](#) - Full AXI4 monitoring (detailed reference)
- [axi_monitor_filtered](#) - Core monitor engine (rtl/amba/monitor/)
- [Monitor Configuration Guide](#) - Configuration strategies

5.10.3 Related Modules

- [axil4_master_wr_mon](#) - Master write with monitoring
- [axil4_slave_rd_mon](#) - Slave read with monitoring

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6 AXIL4 Master Read Monitor (Clock-Gated)

Module: `axil4_master_rd_mon_cg.sv` **Base Module:** [axil4_master_rd_mon](#) **Location:** `rtl/amba/monitor/` **Status:** ⚠ Partial — see [Implementation Status](#)

6.1 Implementation Status

This wrapper gates the monitor’s clock for real. It instantiates `amba_clock_gate_ctrl`, and the base `axil4_master_rd_mon` inside it is driven from `gated_aclk`, not from `aclk`.

What it does:

1. **Gates the clock.** `amba_clock_gate_ctrl` counts `cfg_cg_idle_count` idle cycles and stops `gated_aclk`, reporting on `cg_gating` (the gated clock is stopped) and `cg_idle` (no activity observed).
2. **Holds off the interfaces while gated.** `fub_axil_rready` and `m_axil_rready` are forced low under `cg_gating`, so nothing is accepted into a stopped clock.
3. **Masks the monbus valid while gated** (`monbus_valid = w_monbus_valid && !cg_gating`), which is what makes delivery exactly-once across a gating edge rather than a held valid replayed on wake.

Two earlier versions of this page were wrong in opposite directions, so both are worth naming. One described a `cg_cycles_saved` output, a `cfg_cg_idle_threshold` input and `ENABLE_CLOCK_GATING / CG_IDLE_CYCLES` parameters; none of those exist anywhere in `rtl/amba`. The other – the correction to the first – concluded that the wrapper therefore gates nothing and “will not reduce dynamic power”. That was the more damaging error: it is false, and it steers an integrator away from the right part. The real controls are `cfg_cg_enable` and `cfg_cg_idle_count` (width `CG_IDLE_COUNT_WIDTH`); there is no cycles-saved counter of any kind.

Gating behaviour is asserted directly by `val/amba/test_mon_cg_gating.py` (phase 5 for the ready hold-off, phase 6 for exactly-once monbus delivery).

6.2 Overview

`axil4_master_rd_mon_cg` wraps [axil4_master_rd_mon](#) and adds a power-management control and status interface. All monitoring, filtering, address-range checking, and performance-monitoring behavior is that of the base module; see [axil4_master_rd_mon.md](#) for the complete functional specification.

6.3 Additional Parameters

In addition to all [axil4_master_rd_mon](#) parameters (including `USE_MONITOR` and `N_ADDR_RANGES`):

Parameter	Type	Default	Description
<code>ACLK_MHZ</code>	int	100	Clock frequency in MHz. Builds the microsecond tick LUT in <code>counter_freq_invariant</code> . Leave this at 100 on a 90 MHz part and every us-denominated timeout is wrong, silently – it was unreachable through this wrapper until 2026-09-01.
<code>CFI_MIN_FREQ_MHZ</code>	int	<code>ACLK_MHZ</code>	Lowest frequency the tick LUT must cover (dynamic-frequency builds).
<code>CFI_MAX_FREQ_MHZ</code>	int	<code>ACLK_MHZ</code>	Highest frequency the tick LUT must cover.
<code>USE_WDATA_ORDER_Q</code>	bit	0	Write-data ordering queue. Required (=1) whenever <code>NUM_BANKS > 1</code> .
<code>NUM_BANKS</code>	int	1	Transaction-table

Parameter	Type	Default	Description
			banking. >1 needs USE_WDATA_ORDER_Q=1; the inner module's elaboration guard fires otherwise.
ADDR_FILTER_ENABLE	bit	0	Synthesises the address-range report filter. The parameter only decides whether the logic EXISTS – a build that sets it and leaves <code>cfg_addr_filter_enable</code> low filters nothing and looks broken.
CG_IDLE_COUNT_WIDTH	int	4	Width of <code>cfg_cg_idle_count</code> ; sets the longest programmable idle threshold

There are no CG_GATE_MONITOR, CG_GATE_REPORTER, or CG_GATE_TIMERS parameters, and no independent gating domains. Earlier revisions of this document described such a scheme; it was never implemented.

6.4 Additional Ports

6.4.1 Filter configuration (forwarded)

These reach the inner monitor through this wrapper; before 2026-09-01 they did not.

Port	Direction	Width	Description
<code>cfg_addr_filter_enable</code>	Input	1	High: suppress packets for transactions outside the window. Low: inert, whatever ADDR_FILTER_ENABLE says

Port	Direction	Width	Description
cfg_addr_filter_low	Input	ADDR_WIDT H	Window base, inclusive
cfg_addr_filter_high	Input	ADDR_WIDT H	Window limit, inclusive

There is no runtime ID filter on AXI4-Lite: the protocol has no IDs to match.

All base-module ports are forwarded unchanged, including the `cam_clear` control input (Input, 1) - synchronous clear of the monitor transaction CAM (driven from the harness clear control bit, e.g. CTRL[4]). The wrapper adds:

Port	Direction	Width	Description
cfg_cg_enable	Input	1	Gates the monitor functionally (ANDed into <code>cfg_monitor_enable</code>). 0 = monitor disabled.

The base module's busy output remains available on this wrapper.

6.5 Performance Monitoring

The wrapper forwards the base module's full performance-monitoring interface to `axi_monitor_base` **unchanged** — the power-management interface neither adds, removes, nor retimes any perfmon port. They behave exactly as documented for [axil4_master_rd_mon](#):

- **Config inputs:** `cfg_perf_enable`, `cfg_start_event_sel`, `cfg_end_event_sel`, `cfg_start_trigger`, `cfg_end_trigger`, `cfg_window_force_close`
- **Status / counters:** `window_active`, `window_cycles`, `perf_prod_cycles`, `perf_bp_cycles`, `perf_starv_cycles`, `perf_idle_cycles`, `perf_beat_count`, `perf_byte_count`, `perf_burst_count`

The completion/threshold/debug enables (`cfg_compl_enable`, `cfg_threshold_enable`, `cfg_debug_enable`) and the synthesis-cone parameters (`ENABLE_ERROR_LOGIC`, `ENABLE_TIMEOUT_LOGIC`, `ENABLE_COMPL_LOGIC`, `ENABLE_THRESHOLD_LOGIC`, `ENABLE_PERF_LOGIC`, `ENABLE_DEBUG_LOGIC`) are likewise forwarded unchanged. The utilization buckets watch the **R** (read-data) channel; for AXI4-Lite each transaction is a single data beat, so `perf_burst_count` counts AR handshakes = transactions.

Note that `cfg_cg_enable = 0` disables the monitor, and therefore stops the performance counters as well.

6.6 Usage Example

```
axil4_master_rd_mon_cg #(
    // Base module parameters (see axil4_master_rd_mon.md)
    .AXIL_ADDR_WIDTH(32),
    .AXIL_DATA_WIDTH(32),
    .SKID_DEPTH_AR(2),
    .SKID_DEPTH_R(4),

    // Monitor parameters
    .UNIT_ID(8'h01),
    .AGENT_ID(16'h000A),
    .MAX_TRANSACTIONS(8)
) u_cg (
    .aclk(clk),
    .aresetn(rst_n),

    // Power-management interface
    .cfg_cg_enable(1'b1),           // 0 would disable the monitor
    .cfg_cg_idle_count(4'd4),       // idle cycles before the clock
    stops                             // high while the gated clock is
    .cg_gating(cg_gating),           // high while the gated clock is
    stopped                           // high while the gated clock is
    .cg_idle(cg_idle),

    // ... all other ports same as axil4_master_rd_mon
);
```

6.7 Verification Considerations

Because no clock is actually gated, simulation of this wrapper behaves exactly like the base module as long as `cfg_cg_enable = 1`. Drive `cfg_cg_enable = 1` for any test that expects monitor packets — with it low the monitor is off and no packets are emitted.

6.8 Related Modules

- [axil4_master_rd_mon](#) - Base module (functional specification)

- [axil4_master_wr_mon_cg](#) - Companion monitor wrapper
 - [axi_monitor_base](#) - Core monitoring infrastructure
 - [axi_monitor_filtered](#) - Filtering capabilities
 - [AXIL4 Clock-Gated Variants Guide](#) - The transport-level _cg modules, which do perform real clock gating
-

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6.9 Navigation

- [← Back to Base Module](#)
- [← Back to AXIL4 Index](#)
- [← Back to rtl-amba Index](#)
- [← Back to Main Documentation Index](#)

7 AXIL4 Master Write

Module: axil4_master_wr.sv **Location:** rtl/amba/axil4/ **Status:** ✓ Production Ready

7.1 Overview

The AXIL4 Master Write module provides a buffered AXI4-Lite write interface for master devices. AXI4-Lite is a simplified subset of AXI4 designed for control register access with reduced signal count and single-beat transactions only.

7.1.1 Key Features

- ✓ **AXI4-Lite Protocol:** Simplified write-only interface (AW, W, B channels)
 - ✓ **Single-Beat Transactions:** No burst support (always 1 transfer)
 - ✓ **Buffered Channels:** Configurable skid buffers for all 3 channels
 - ✓ **Byte Strobes:** WSTRB support for partial word writes
 - ✓ **Elastic Buffering:** Decouples frontend and backend timing
 - ✓ **Activity Monitoring:** Busy signal for clock gating
 - ✓ **Minimal Latency:** 1-cycle buffer overhead per channel
-

7.2 Parameters

Parameter	Type	Default	Description
AXIL_ADDR_WIDTH	int	32	Address bus width (typically 32)
AXIL_DATA_WIDTH	int	32	Data bus width (32 or 64)
SKID_DEPTH_AW	int	2	AW channel skid buffer depth, in entries
SKID_DEPTH_W	int	2	W channel skid buffer depth, in entries
SKID_DEPTH_B	int	2	B channel skid buffer depth, in entries

7.2.1 Derived Parameters

Each channel's skid buffer stores the whole channel payload as one packed vector, so the derived *Size parameters are just the sum of the widths of the signals concatenated into that buffer.

Parameter	Calculation	Description
AW	AXIL_ADDR_WIDTH	Internal address width alias
DW	AXIL_DATA_WIDTH	Internal data width alias
AWSize	AW+3	AW packet: {awaddr[AW-1:0], awprot[2:0]} — the 3 is AWPROT
WSize	DW+(DW/8)	W packet: {wdata[DW-1:0], wstrb[DW/8-1:0]} — one strobe bit per byte lane
BSize	2	B packet: bresp[1:0] only

7.3 Port Groups

7.3.1 Clock and Reset

Port	Direction	Width	Description
aclk	Input	1	AXI clock
aresetn	Input	1	Active-low asynchronous reset

7.3.2 Frontend AXI4-Lite Write Interface (Input Side)

Write Address Channel (AW): | Port | Direction | Width | Description | |——|———|——|
———| | fub_awaddr | Input | AW | Write address | | fub_awprot | Input | 3 | Protection attributes | | fub_awvalid | Input | 1 | Address valid | | fub_awready | Output | 1 | Address ready (from buffer) |

Write Data Channel (W): | Port | Direction | Width | Description | |——|———|——|
———| | fub_wdata | Input | DW | Write data | | fub_wstrb | Input | DW/8 | Write strobes (byte lane enables) | | fub_wvalid | Input | 1 | Data valid | | fub_wready | Output | 1 | Data ready (from buffer) |

Write Response Channel (B): | Port | Direction | Width | Description | |——|———|——|
———| | fub_bresp | Output | 2 | Write response (OKAY, SLVERR, DECERR) | | fub_bvalid | Output | 1 | Response valid (from buffer) | | fub_bready | Input | 1 | Response ready |

7.3.3 Backend AXI4-Lite Master Interface (Output Side)

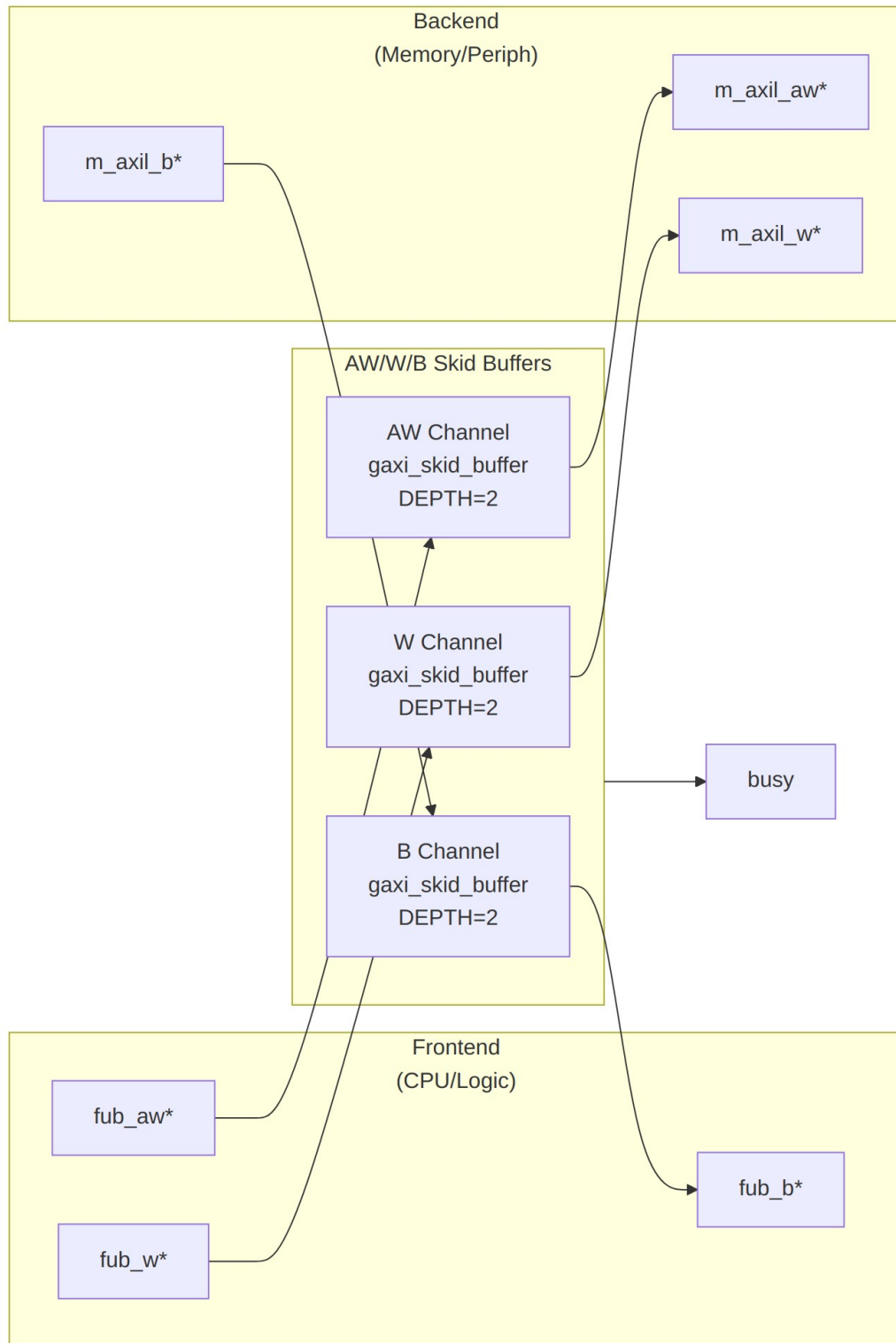
Write Address Channel (AW): | Port | Direction | Width | Description | |——|———|——|
———| | m_axil_awaddr | Output | AW | Write address (to slave) | | m_axil_awprot | Output | 3 | Protection attributes | | m_axil_awvalid | Output | 1 | Address valid (from buffer) | | m_axil_awready | Input | 1 | Address ready (from slave) |

Write Data Channel (W): | Port | Direction | Width | Description | |——|———|——|
———| | m_axil_wdata | Output | DW | Write data (to slave) | | m_axil_wstrb | Output | DW/8 | Write strobes | | m_axil_wvalid | Output | 1 | Data valid (from buffer) | | m_axil_wready | Input | 1 | Data ready (from slave) |

Write Response Channel (B): | Port | Direction | Width | Description | |——|———|——|
———| | m_axil_bresp | Input | 2 | Write response (from slave) | | m_axil_bvalid | Input | 1 | Response valid (from slave) | | m_axil_bready | Output | 1 | Response ready (to slave) |

7.3.4 Status Outputs

Port	Direction	Width	Description
busy	Output	1	Interface active (for clock gating)



Module Architecture

7.4 Signal Flow

7.4.1 Write Transaction Sequence

Cycle 1: fub_awaddr=0x2000, fub_awvalid=1, fub_awprot=0
fub_wdata=0xCAFEBAFE, fub_wstrb=4'b1111, fub_wvalid=1
fub_awready=1, fub_wready=1 (buffers accept)

Cycle 2: Buffers forward to backend:
m_axil_awaddr=0x2000, m_axil_awvalid=1
m_axil_wdata=0xCAFEBAFE, m_axil_wstrb=4'b1111,
m_axil_wvalid=1
m_axil_awready=1, m_axil_wready=1 (slave accepts)

Cycle 3-5: Slave processes write

Cycle 6: m_axil_bresp=OKAY, m_axil_bvalid=1
m_axil_bready=1 (B buffer accepts)

Cycle 7: B buffer forwards to frontend:
fub_bresp=OKAY, fub_bvalid=1
fub_bready=1 (frontend accepts)

7.5 Usage Example

7.5.1 Basic Integration

```
axil4_master_wr #(
    .AXIL_ADDR_WIDTH(32),
    .AXIL_DATA_WIDTH(32),
    .SKID_DEPTH_AW(2),      // 2 entries
    .SKID_DEPTH_W(2),       // 2 entries
    .SKID_DEPTH_B(2)        // 2 entries
) u_axil_master_wr (
    .aclk          (axi_clk),
    .aresetn       (axi_resetn),

    // Frontend interface (from CPU)
    .fub_awaddr    (cpu_awaddr),
    .fub_awprot    (cpu_awprot),
    .fub_awvalid   (cpu_awvalid),
    .fub_awready   (cpu_awready),
```

```

.fub_wdata      (cpu_wdata),
.fub_wstrb      (cpu_wstrb),
.fub_wvalid     (cpu_wvalid),
.fub_wready     (cpu_wready),

.fub_bresp      (cpu_bresp),
.fub_bvalid     (cpu_bvalid),
.fub_bready     (cpu_bready),

// Backend interface (to peripheral/memory)
.m_axil_awaddr  (periph_awaddr),
.m_axil_awprot  (periph_awprot),
.m_axil_awvalid (periph_awvalid),
.m_axil_awready (periph_awready),

.m_axil_wdata   (periph_wdata),
.m_axil_wstrb   (periph_wstrb),
.m_axil_wvalid  (periph_wvalid),
.m_axil_wready  (periph_wready),

.m_axil_bresp   (periph_bresp),
.m_axil_bvalid  (periph_bvalid),
.m_axil_bready  (periph_bready),

// Status
.busy           (wr_busy)
);

```

7.5.2 Register Write Example (Testbench Stimulus)

The two snippets below are **testbench stimulus, not RTL**. They use an `initial` block and blocking assignments to drive the frontend ports and are not synthesizable. In a real design the frontend handshake is driven by an `always_ff` block or by the register-access state machine of the requesting logic.

```

// Write to control register at 0x2000
initial begin
    @(posedge aclk);

    // Present address and data simultaneously
    fub_awaddr  = 32'h2000;
    fub_awprot  = 3'b000;
    fub_awvalid = 1'b1;

    fub_wdata   = 32'hDEADBEEF;

```

```

fub_wstrb    = 4'b1111; // All bytes valid
fub_wvalid   = 1'b1;

@(posedge aclk);
while (!(fub_awready && fub_wready)) @(posedge aclk);
fub_awvalid  = 1'b0;
fub_wvalid   = 1'b0;

// Wait for write response
fub_bready   = 1'b1;
@(posedge aclk);
while (!fub_bvalid) @(posedge aclk);

$display("Write response: BRESP=%0d", fub_bresp);
end

```

7.5.3 Partial Write Example (Byte Strobes)

```

// Write only upper 16 bits to 32-bit register
fub_awaddr   = 32'h2004;
fub_awprot   = 3'b000;
fub_awvalid  = 1'b1;

fub_wdata    = 32'hABCD0000; // Data in upper half
fub_wstrb    = 4'b1100;      // Only upper 2 bytes valid
fub_wvalid   = 1'b1;

// Result: Register[31:16] = 0xABCD, Register[15:0] unchanged

```

7.6 Design Notes

7.6.1 AXI4-Lite Protocol Constraints

Single-Beat Only: - No burst support (implicitly AWLEN=0, AWSIZE=log2(DW/8)) - Each AW+W transaction results in exactly 1 B response - No WLAST signal (always last and only beat)

No Out-of-Order: - No AWID/BID signals - Responses always in order - AW and W can arrive in any order as far as **this module** is concerned: the AW and W skid buffers are fully independent and nothing here pairs an address with its data. The protocol permits either order, but a downstream slave may still require AW before W (or at least in the same cycle) to decode the target before it can consume the data. Check the slave's requirements — this module will not reorder for you.

Reduced Signals: - No AWCACHE, AWQOS, AWREGION, AWLOCK, AWSIZE, AWLEN, AWBURST - Only AWADDR and AWPROT for control

7.6.2 Buffer Depth Selection

SKID_DEPTH_AW, SKID_DEPTH_W, and SKID_DEPTH_B are passed straight through to the DEPTH parameter of gaxi_skid_buffer. DEPTH is the **literal entry count**, not an exponent, and is constrained to one of 2..8 inclusive.

AW Channel (Addresses): - SKID_DEPTH_AW = 2 (2 entries): Typical for control registers - Increase if many back-to-back writes with slow slave

W Channel (Data): - SKID_DEPTH_W = 2 (2 entries): Matches AW depth - Should be >= SKID_DEPTH_AW to avoid W channel stalls

B Channel (Responses): - SKID_DEPTH_B = 2 (2 entries): Responses are single-beat - Smaller depth acceptable if frontend can always accept

7.6.3 Write Strobe (WSTRB) Usage

32-bit Data Width: |WSTRB[3:0] | Bytes Written | Use Case | |———|———|———| |
4'b0001 | [7:0] | Byte write to offset 0 | | 4'b0010 | [15:8] | Byte write to offset 1 | | 4'b0011 |
[15:0] | Half-word write | | 4'b1100 | [31:16] | Upper half-word | | 4'b1111 | [31:0] | Full word
write |

64-bit Data Width: |WSTRB[7:0] | Bytes Written | Use Case | |———|———|———| |
8'b0000_0001 | [7:0] | Byte write | | 8'b0000_1111 | [31:0] | Lower word | | 8'b1111_0000 |
[63:32] | Upper word | | 8'b1111_1111 | [63:0] | Full double-word |

7.6.4 Busy Signal

busy is the OR of three persistent terms and two transient terms:

Term	Kind	Condition
AW buffer occupancy	Persistent	int_aw_count > 0
W buffer occupancy	Persistent	int_w_count > 0
B buffer occupancy	Persistent	int_b_count > 0
Frontend presenting a new transaction	Transient	fub_awvalid fub_wvalid
Backend presenting a response	Transient	m_axil_bvalid

The two transient terms can be high for a single cycle, so busy itself can pulse. This is intentional: the clock-gate controller does not gate on busy directly. amba_clock_gate_ctrl registers

activity into a wakeup flop and then requires `cfg_cg_idle_count` consecutive idle cycles before gating, so a one-cycle handshake reloads the idle counter rather than causing the gate to oscillate.

Use for: - Clock gating control (see `axil4_master_wr_cg`) - Power management - Interface idle detection

7.7 Performance Characteristics

7.7.1 Latency

Path	Cycles	Notes
Frontend → Backend (AW)	1	Skid buffer overhead
Frontend → Backend (W)	1	Skid buffer overhead
Backend → Frontend (B)	1	Skid buffer overhead
Total write latency	Slave latency + 3	AW + W + Slave + B, best case

The +3 best case assumes the slave asserts `AWREADY` and `WREADY` in the **same** cycle, so the AW and W buffer traversals overlap. If the slave accepts the address and the data in different cycles, the write completes off the later of the two handshakes, and the total is that skew plus the slave and B latency. Backpressure on B adds further cycles.

7.7.2 Throughput

Scenario	Rate	Notes
Back-to-back writes	1 write/cycle	If buffers not stalled
Typical peripheral	1 write/N cycles	N = slave response time

7.7.3 Resource Usage

These are **order-of-magnitude estimates**, not measured synthesis results — no target device, tool version, or optimization setting is attached to them. They scale with `AXIL_DATA_WIDTH` and the `SKID_DEPTH_*` settings. Synthesize for your own device before budgeting area.

Resource	Count	Notes
LUTs	~300	Estimate, 32-bit data, 3 channels, default

Resource	Count	Notes
		depths
FFs	~250	Buffer state + control
BRAM	0	Skid buffers are flop-based

7.8 Related Modules

7.8.1 Core Modules

- [axil4_master_rd](#) - Read master counterpart
- [axil4_slave_rd](#) - Slave read interface
- [axil4_slave_wr](#) - Slave write interface

7.8.2 Monitor Modules

- [axil4_master_wr_mon](#) - Master write with monitoring (rtl/amba/monitor/)

7.8.3 Clock-Gated Variants

- [axil4_master_wr_cg](#) - Clock-gated version

7.8.4 Supporting Infrastructure

- [gaxi_skid_buffer](#) - Elastic buffering (rtl/amba/gaxi/)
-

7.9 References

7.9.1 Specifications

- ARM IHI 0022E: AMBA AXI and ACE Protocol Specification
- Chapter 4: AXI4-Lite (simplified subset)

7.9.2 Documentation

- [rtl-amba Overview](#) - Complete AMBA subsystem
- [AXI4 Master Write](#) - Full AXI4 comparison

7.9.3 Source Code

- RTL: rtl/amba/axil4/axil4_master_wr.sv

- Tests: val/amba/test_axil4_master_wr.py
 - Framework: bin/TBClasses/components/axil4/
-

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8 AXIL4 Master Write Interface (Clock-Gated)

Module: axil4_master_wr_cg.sv **Base Module:** [axil4_master_wr](#) **Location:** rtl/amba/axil4/ **Status:** ✓ Production Ready

8.1 Quick Reference

This is the **clock-gated** variant of [axil4_master_wr](#).

For complete clock-gating documentation, usage examples, and configuration guidelines, see:

- [AXIL4 Clock-Gated Variants Guide](#) (AXIL4-specific)
 - [Clock-Gated Variants Guide](#) (cross-protocol overview)
-

8.2 Summary

The axil4_master_wr_cg module adds power optimization to axil4_master_wr through activity-based clock gating:

- ✓ **Same Functionality:** 100% equivalent to base module
- ✓ **Power Savings:** dynamic clock power in the gated block scales with idle fraction; see the [AXIL4 Clock-Gated Variants Guide](#) for the estimate table and its caveats

- ✓ **Configurable:** idle threshold and enable, both at **runtime** via `cfg_cg_idle_count` / `cfg_cg_enable`
- ✓ **Zero Overhead When Disabled:** `cfg_cg_enable = 0` holds the clock free-running, making behavior identical to the base module

8.3 Common Parameters

In addition to all `axil4_master_wr` parameters:

The `_cg` wrapper adds exactly **one** parameter. Gating is enabled and tuned at runtime through ports, not through parameters — there is no `ENABLE_CLOCK_GATING` parameter, no `CG_IDLE_CYCLES` parameter, and no per-domain `CG_GATE_*` parameters on this module.

Parameter	Type	Default	Description
<code>CG_IDLE_COUNT_WIDTH</code>	int	4	Width of the idle counter; max idle count = $2^N - 1$

8.3.1 Additional Ports

Port	Direction	Width	Description
<code>cfg_cg_enable</code>	Input	1	Enable clock gating (0 = clock always running)
<code>cfg_cg_idle_count</code>	Input	<code>CG_IDLE_COUNT_WIDTH</code>	Idle cycles before gating
<code>cg_gating</code>	Output	1	Clock currently gated
<code>cg_idle</code>	Output	1	No activity on the previous cycle

The base module's busy output is **not** exposed on the `_cg` wrapper; it is consumed internally as a wakeup term. All other ports are identical to `axil4_master_wr`.

8.4 Quick Usage

```
axil4_master_wr_cg #(
    // Base module parameters (see axil4_master_wr.md)
    .AXIL_ADDR_WIDTH(32),
    .AXIL_DATA_WIDTH(32),
    .SKID_DEPTH_AW(2),
```

```

        .SKID_DEPTH_W(2),
        .SKID_DEPTH_B(2),

        // Clock gating (see CG guide for details)
        .CG_IDLE_COUNT_WIDTH(4)
    ) u_cg (
        .aclk(clk),
        .aresetn(rst_n),

        // Clock gating control (runtime, not parameters)
        .cfg_cg_enable(1'b1),
        .cfg_cg_idle_count(4'd5),

        // ... all other ports same as axil4_master_wr, except `busy`,
        // which is replaced by cg_gating / cg_idle
        .cg_gating(clk_gated),
        .cg_idle(if_idle)
    );

```

8.5 Documentation

- **Base Module Functionality:** [axil4_master_wr.md](#)
 - **AXIL4 Clock Gating Guide:** [axil4_clock_gating_guide.md](#)
 - **Cross-Protocol Clock Gating Guide:** [clock_gated_variants.md](#)
 - **Detailed CG Examples:**
 - [axi4_master_rd_mon_cg.md](#) (AXI4 monitor)
 - [axil4_master_rd_mon_cg.md](#) (AXIL4 monitor)
 - [apb4_slave_cg.md](#) (APB interface)
-

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9 AXIL4 Master Write with Monitoring

Module: axil4_master_wr_mon.sv **Location:** rtl/amba/monitor/ **Status:** ✓ Production Ready

9.1 Overview

Combines [axil4_master_wr](#) with [axi_monitor_filtered](#) for write transaction monitoring.

9.1.1 Key Features

- ✓ All features of [axil4_master_wr](#)
 - ✓ Integrated write monitoring (AW, W, B channels)
 - 2-level filtering (packet-type masks, then per-event-code masks) and error detection
 - ✓ Simplified for AXI4-Lite (MAX_TRANSACTIONS=8)
-

9.2 Additional Parameters

Identical to [axil4_master_rd_mon](#) including: - UNIT_ID, AGENT_ID, MAX_TRANSACTIONS, ENABLE_FILTERING, ADD_PIPELINE_STAGE, USE_MONITOR, N_ADDR_RANGES - ACLK_MHZ (default 100) and CFI_MIN_FREQ_MHZ / CFI_MAX_FREQ_MHZ (default ACLK_MHZ) – the microsecond tick LUT. **Leave ACLK_MHZ at 100 on a 90 MHz part and every us-denominated timeout is wrong, silently** - USE_WDATA_ORDER_Q (default 0) and NUM_BANKS (default 1) – **NUM_BANKS > 1 on a WRITE monitor requires USE_WDATA_ORDER_Q=1**, or axi_monitor_trans_mgr fails elaboration - ADDR_FILTER_ENABLE (default 0) – synthesises the address-range report filter; the cfg_addr_filter_* ports arm it at runtime - ACTIVE_TRANS_THRESHOLD (default MAX_TRANSACTIONS/2): active-transaction count that trips a threshold packet when cfg_threshold_enable=1; replaces the former hardwired value - The synthesis-cone parameters ENABLE_ERROR_LOGIC, ENABLE_TIMEOUT_LOGIC, ENABLE_COMPL_LOGIC, ENABLE_THRESHOLD_LOGIC, ENABLE_PERF_LOGIC (all default 1) and ENABLE_DEBUG_LOGIC (default 0), each of which drops the matching detection cone when set to 0. ENABLE_PERF_PACKETS is tied 1'b1 and ENABLE_DEBUG_MODULE 1'b0 internally.

See [axil4_master_rd_mon](#) for complete parameter descriptions. The removed CAM_PIPELINE / TRANS_CAM_PIPELINE parameters no longer exist (the CAM is always pipelined).

9.3 Performance Monitoring

When performance monitoring is enabled, the wrapper forwards a **measurement-window state machine** plus a bank of W-channel (write-data) utilization counters to `axi_monitor_base`. All counters accumulate **only while a window is open** (`window_active = 1`) and hold their values between windows so the host can read a completed window's totals. The counters advance only while `cfg_perf_enable = 1`. `ENABLE_PERF_LOGIC = 0` does NOT drop them: the window FSM and its counters are unconditional always_ff blocks in `axi_monitor_base`, outside every generate. That parameter gates only `g_perf` in the reporter – the legacy perf-packet cone and the two lifetime counters. `USE_MONITOR = 0` is what ties the perfmon outputs off.

Avoid enabling completion (`cfg_compl_enable`) and performance (`cfg_perf_enable`) packets simultaneously under heavy traffic — the monitor bus sustains at most one packet per two cycles. Runtime-disabling either class is safe (terminal entries auto-retire; see [axi_monitor_reporter](#)); alternatively, `cfg_axi_pkt_mask` drops the packets while keeping marking and counting. See [docs/user-guides/AXI_Monitor_Configuration_Guide.md](#).

9.3.1 The Measurement Window

A window is opened by a **start event** and closed by an **end event**:

- `cfg_start_event_sel / cfg_end_event_sel` (3-bit) select the event source (e.g. `3'b010` selects the `cfg_perf_enable` edge).
- `cfg_start_trigger / cfg_end_trigger` pulses fire the window directly from an engine or CSR.
- `cfg_window_force_close` is a software override that closes the window immediately.

While the window is open, `window_active` is high and `window_cycles` [31:0] free-runs, counting every clock elapsed inside the window.

9.3.2 Utilization Counters (W channel)

Every in-window cycle is classified by the W-channel valid/ready into exactly one of four buckets:

Output	Width	Condition	Meaning
<code>perf_prod_cycles</code>	32	<code>wvalid && wready</code>	productive beat transferred
<code>perf_bp_cycles</code>	32	<code>wvalid && !wready</code>	back-pressure (data offered, consumer not

Output	Width	Condition	Meaning
perf_starv_cycles	32	!wvalid && wready	ready) starvation (consumer ready, no valid data)
perf_idle_cycles	32	!wvalid && !wready	idle

The four buckets sum to window_cycles, so utilization = perf_prod_cycles / window_cycles.

9.3.3 Throughput Counters

Output	Width	Meaning
perf_beat_count	32	data beats transferred (= perf_prod_cycles, 1 beat/cycle)
perf_byte_count	64	bytes transferred = beats × (1 << AxSIZE); AXIL is fixed at AxSIZE=3'b010 (4 bytes)
perf_burst_count	32	AW (write-address) handshakes

AXI4-Lite note: every transaction is a single data beat (AWLEN is implicitly 0), so perf_burst_count counts AW handshakes = transactions and perf_beat_count equals the transaction count. Average burst length (perf_beat_count / perf_burst_count) is therefore always 1.

9.3.4 Perfmon & Additional Control Ports

The wrapper adds the perfmon config/status ports (cfg_start_event_sel, cfg_end_event_sel, cfg_start_trigger, cfg_end_trigger, cfg_window_force_close, window_active, window_cycles, perf_prod_cycles, perf_bp_cycles, perf_starv_cycles, perf_idle_cycles, perf_beat_count, perf_byte_count, perf_burst_count) and the completion/threshold/debug enables (cfg_compl_enable, cfg_threshold_enable, cfg_debug_enable) with the same directions/widths and semantics as

[axil4_master_rd_mon](#) — the only difference is that perf_burst_count counts AW handshakes and the utilization buckets watch the W channel. As on the read monitor, cfg_debug_level/cfg_debug_mask/cfg_active_trans_threshold are tied to constants internally and not exposed.

9.4 Monitor Backpressure (block_ready)

block_ready is an internal flow-control net inside the wrapper – it is not a port. It goes low when the monitor’s transaction-table occupancy reaches its blocking threshold (a function of MAX_TRANSACTIONS; the reporter FIFO depth INTR_FIFO_DEPTH has no path to it). The wrapper ANDs it into the upstream-facing fub_axil_aware so a saturated monitor throttles new transactions at the handshake instead of dropping events.

- **Where the stall lands:** the upstream fub_axil_aware is forced low until the monitor drains.
- **When USE_MONITOR=0:** block_ready is internally tied high, so the wrapper imposes no stall and runs at full bandwidth.
- **When cfg_monitor_enable=0:** the wrapper gate forces the upstream ready open, so a runtime-disabled monitor can never stall the datapath (the AXI4 monitors assert this as an in-RTL formal property, ap_disabled_never_stalls; this module has no ifdef FORMAL block of its own, so here the guarantee rests on the gate expression above, not a proof).

Recovery is guaranteed by the **saturation-recovery contract**: command-originated table entries are capped at $\text{MAX_TRANSACTIONS} - \text{cmd_entry_reserve}(\text{MAX_TRANSACTIONS})$ (reserve = 4 for tables of 16 or more, 0 below; the function lives in monitor_common_pkg), and block_ready re-asserts at occupancy $< \text{MAX_TRANSACTIONS} - (\text{reserve} - 1)$ – a threshold strictly ABOVE the command cap – so a saturated table always drains back below the reopen point. Blocking throttles; it never deadlocks. Tables smaller than 16 keep full legacy allocation (small tables cannot spare slots) and trade the recovery guarantee for tracking capacity. The contract is verified by in-RTL formal properties (mutation-checked) and a 100-seed deliberately-undersized-table stream sweep; see [axi_monitor_base](#) for the canonical description.

Sizing note: a monitor on a bus shared by several channels/requesters must size MAX_TRANSACTIONS to cover $\text{NUM_CHANNELS} \times \text{per-channel outstanding}$ plus margin – the per-channel limit alone makes the monitor throttle the shared master. Tables deeper than 64 also need Verilator’s `--unroll-count` raised (default 64) in sim builds.

9.5 Address-Range Checker

Identical to [axil4_master_rd_mon](#) except the checker watches AW (write address) handshakes instead of AR (read address) handshakes. The `cfg_addr_*` configuration inputs and monbus event encoding are the same. See the read monitor’s Address-Range Checker section for full details.

9.5.1 Packet filters

Two independent runtime filters cut monbus traffic without touching the datapath. Both are inert until driven, which is the failure to watch for: the build-time parameter only decides whether the logic is SYNTHESISED, and a design that sets the parameter but leaves the `cfg_*` ports tied low filters nothing and looks like the feature is broken.

Address-range filter — `ADDR_FILTER_ENABLE` (bit, default 0) builds it.

Port	Width	Description
<code>cfg_addr_filter_enable</code>	1	High: suppress packets for transactions outside the window. Low: inert, whatever the parameter says
<code>cfg_addr_filter_low</code>	<code>ADDR_WIDTH</code>	Window base, inclusive
<code>cfg_addr_filter_high</code>	<code>ADDR_WIDTH</code>	Window limit, inclusive

The verdict is latched per table entry at `ALLOCATION`, from the command address, and held for that entry's life. Widening the window mid-flight does not un-filter entries already admitted — which is what makes it safe to reprogram live. Contrast the address-range `CHECKER` above, which re-evaluates per command.

There is no runtime ID filter here. AXI-Lite has no transaction IDs, so this wrapper ties `cfg_id_filter_enable` / `cfg_id_match_base` / `cfg_id_match_count` off on the inner monitor rather than exposing them — a filter keyed on a field the protocol lacks has nothing to match against.

9.6 Additional Ports

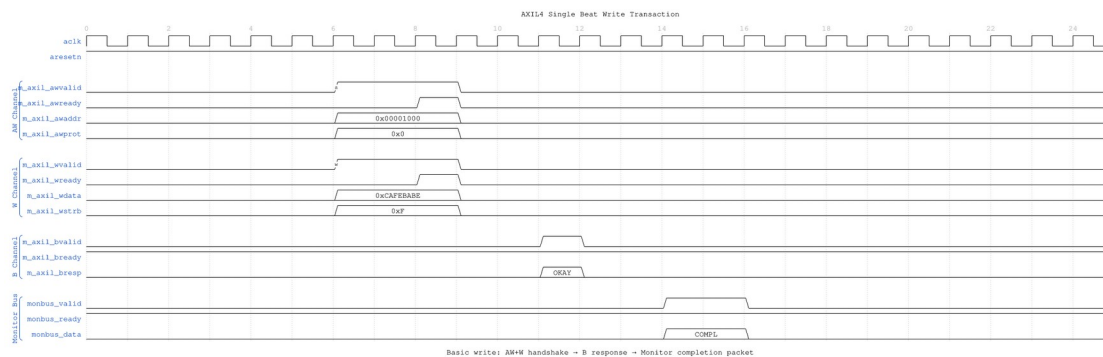
Same as [axil4_master_rd_mon](#): - Monitor configuration inputs, including `cfg_error_enable`, `cfg_timeout_enable`, `cfg_compl_enable`, `cfg_threshold_enable`, `cfg_perf_enable`, `cfg_debug_enable` - `cam_clear` control input (Input, 1) - synchronous clear of the monitor transaction CAM (driven from the harness clear control bit, e.g. `CTRL[4]`) - Filtering masks (7 masks) - The performance-monitoring config/status ports (see the [Performance Monitoring](#) section above and the [read-monitor port table](#)) - `cfg_freq_sel` (Input, 4) - `counter_freq_invariant` LUT index scaling the 1 us timer tick; the microsecond timeouts are measured in these ticks. With the default `CFI_MIN_FREQ_MHZ == CFI_MAX_FREQ_MHZ == ACLK_MHZ` every LUT entry is identical, so it has no effect until you give CFI a real MIN..MAX range - Monitor bus output (valid/ready/data) - Status outputs (busy, active_transactions, error_count)

9.7 Usage

```
axil4_master_wr_mon #(
    .AXIL_ADDR_WIDTH(32),
    .AXIL_DATA_WIDTH(32),
    .UNIT_ID(1),
    .AGENT_ID(11), // Different from read monitor
    .MAX_TRANSACTIONS(8)
) u_axil_wr_mon (
    // AXIL write interfaces (AW, W, B)
    // Monitor configuration and bus
    // Same pattern as axil4_master_rd_mon
);
```

9.8 Timing Diagrams

9.8.1 Scenario 1: Single-Beat Write Transaction

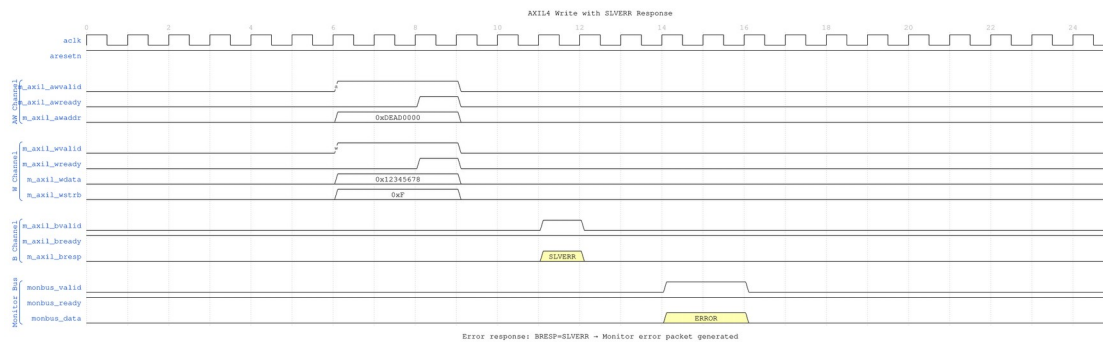


Single Beat Write

WaveJSON: [single_beat_write_001.json](#)

Key Observations: - AW and W channels handshake simultaneously (common in AXIL) - B channel response: Slave returns BRESP=OKAY - Monitor generates completion packet when B phase completes - WSTRB indicates which byte lanes are valid

9.8.2 Scenario 2: Write Error (SLVERR)

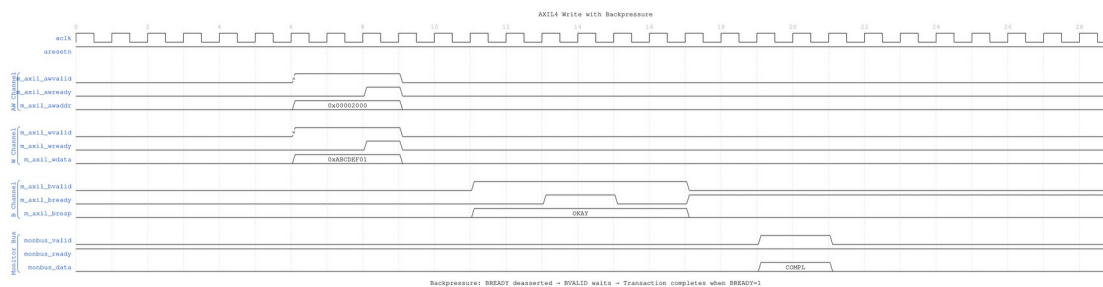


Write Error SLVERR

WaveJSON: [write_error_slvrr_001.json](#)

Key Observations: - Invalid address triggers BRESP=SLVERR - Monitor detects error response and generates ERROR packet - Write data sent but not committed by slave - Error packet includes address and response code

9.8.3 Scenario 3: Write with Backpressure



Write Backpressure

WaveJSON: [write_backpressure_001.json](#)

Key Observations: - Master not ready: BREADY deasserted - Slave holds BVALID until BREADY=1 - Monitor tracks extended write latency - Completion packet generated after B handshake

9.9 Related Modules

- [axil4_master_wr](#) - Base functional module
- [axil4_master_rd_mon](#) - Read monitor counterpart
- [AXI4 Master Write Mon](#) - Full AXI4 reference

10 AXIL4 Master Write Monitor (Clock-Gated)

Module: axil4_master_wr_mon_cg.sv **Base Module:** axil4_master_wr_mon **Location:** rtl/amba/monitor/ **Status:** ⚠ Partial — see [Implementation Status](#)

10.1 Implementation Status

This wrapper gates the monitor’s clock for real. It instantiates `amba_clock_gate_ctrl`, and the base `axil4_master_wr_mon` inside it is driven from `gated_aclk`, not from `aclk`.

What it does:

1. **Gates the clock.** `amba_clock_gate_ctrl` counts `cfg_cg_idle_count` idle cycles and stops `gated_aclk`, reporting on `cg_gating` (the gated clock is stopped) and `cg_idle` (no activity observed).
2. **Holds off the interfaces while gated.** `fub_axil_awready`, `fub_axil_wready` and `m_axil_bready` are forced low under `cg_gating`, so nothing is accepted into a stopped clock.
3. **Masks the monbus valid while gated** (`monbus_valid = w_monbus_valid && !cg_gating`), which is what makes delivery exactly-once across a gating edge rather than a held valid replayed on wake.

Two earlier versions of this page were wrong in opposite directions, so both are worth naming. One described a `cg_cycles_saved` output, a `cfg_cg_idle_threshold` input and `ENABLE_CLOCK_GATING / CG_IDLE_CYCLES` parameters; none of those exist anywhere in `rtl/amba`. The other – the correction to the first – concluded that the wrapper therefore gates nothing and “will not reduce dynamic power”. That was the more damaging error: it is false, and it steers an integrator away from the right part. The real controls are `cfg_cg_enable` and `cfg_cg_idle_count` (width `CG_IDLE_COUNT_WIDTH`); there is no cycles-saved counter of any kind.

Gating behaviour is asserted directly by `val/amba/test_mon_cg_gating.py` (phase 5 for the ready hold-off, phase 6 for exactly-once monbus delivery).

10.2 Overview

`axil4_master_wr_mon_cg` wraps [axil4_master_wr_mon](#) and adds a power-management control and status interface. All monitoring, filtering, address-range checking, and performance-monitoring behavior is that of the base module; see [axil4_master_wr_mon.md](#) for the complete functional specification.

10.3 Additional Parameters

In addition to all [axil4_master_wr_mon](#) parameters (including `USE_MONITOR` and `N_ADDR_RANGES`):

Parameter	Type	Default	Description
<code>ACLK_MHZ</code>	<code>int</code>	100	Clock frequency in MHz. Builds the microsecond tick LUT in <code>counter_freq_invariant</code> . Leave this at 100 on a 90 MHz part and every us-denominated timeout is wrong, silently – it was unreachable through this wrapper until 2026-09-01.
<code>CFI_MIN_FREQ_MHZ</code>	<code>int</code>	<code>ACLK_MHZ</code>	Lowest frequency the tick LUT must cover (dynamic-frequency builds).
<code>CFI_MAX_FREQ_MHZ</code>	<code>int</code>	<code>ACLK_MHZ</code>	Highest frequency the tick LUT must cover.
<code>USE_WDATA_ORDER_Q</code>	<code>bit</code>	0	Write-data ordering queue. Required (=1) whenever <code>NUM_BANKS > 1</code> .
<code>NUM_BANKS</code>	<code>int</code>	1	Transaction-table banking. >1 needs <code>USE_WDATA_ORDER_Q=1</code> ;

Parameter	Type	Default	Description
ADDR_FILTER_ENABLE	bit	0	the inner module's elaboration guard fires otherwise. Synthesises the address-range report filter. The parameter only decides whether the logic EXISTS – a build that sets it and leaves <code>cfg_addr_filter_enable</code> low filters nothing and looks broken.
CG_IDLE_COUNT_WIDTH	int	4	Width of <code>cfg_cg_idle_count</code> ; sets the longest programmable idle threshold

There are no `CG_GATE_MONITOR`, `CG_GATE_REPORTER`, or `CG_GATE_TIMERS` parameters, and no independent gating domains. Earlier revisions of this document described such a scheme; it was never implemented.

10.4 Additional Ports

10.4.1 Filter configuration (forwarded)

These reach the inner monitor through this wrapper; before 2026-09-01 they did not.

Port	Direction	Width	Description
<code>cfg_addr_filter_enable</code>	Input	1	High: suppress packets for transactions outside the window. Low: inert, whatever <code>ADDR_FILTER_ENABLE</code> says
<code>cfg_addr_filter_low</code>	Input	<code>ADDR_WIDTH</code>	Window base, inclusive

Port	Direction	Width	Description
cfg_addr_filter_high	Input	ADDR_WIDT H	Window limit, inclusive

There is no runtime ID filter on AXI4-Lite: the protocol has no IDs to match.

All base-module ports are forwarded unchanged, including the `cam_clear` control input (Input, 1) - synchronous clear of the monitor transaction CAM (driven from the harness clear control bit, e.g. CTRL[4]). The wrapper adds:

Port	Direction	Width	Description
cfg_cg_enable	Input	1	Gates the monitor functionally (ANDed into <code>cfg_monitor_enable</code>). 0 = monitor disabled.

The base module's busy output remains available on this wrapper.

10.5 Performance Monitoring

The wrapper forwards the base module's full performance-monitoring interface to `axi_monitor_base` **unchanged** — the power-management interface neither adds, removes, nor retimes any perfmon port. They behave exactly as documented for [axil4_master_wr_mon](#):

- **Config inputs:** `cfg_perf_enable`, `cfg_start_event_sel`, `cfg_end_event_sel`, `cfg_start_trigger`, `cfg_end_trigger`, `cfg_window_force_close`
- **Status / counters:** `window_active`, `window_cycles`, `perf_prod_cycles`, `perf_bp_cycles`, `perf_starv_cycles`, `perf_idle_cycles`, `perf_beat_count`, `perf_byte_count`, `perf_burst_count`

The completion/threshold/debug enables (`cfg_compl_enable`, `cfg_threshold_enable`, `cfg_debug_enable`) and the synthesis-cone parameters (`ENABLE_ERROR_LOGIC`, `ENABLE_TIMEOUT_LOGIC`, `ENABLE_COMPL_LOGIC`, `ENABLE_THRESHOLD_LOGIC`, `ENABLE_PERF_LOGIC`, `ENABLE_DEBUG_LOGIC`) are likewise forwarded unchanged. The utilization buckets watch the **W** (write-data) channel; for AXI4-Lite each transaction is a single data beat, so `perf_burst_count` counts AW handshakes = transactions.

Note that `cfg_cg_enable = 0` disables the monitor, and therefore stops the performance counters as well.

10.6 Usage Example

```
axil4_master_wr_mon_cg #(
    // Base module parameters (see axil4_master_wr_mon.md)
    .AXIL_ADDR_WIDTH(32),
    .AXIL_DATA_WIDTH(32),
    .SKID_DEPTH_AW(2),
    .SKID_DEPTH_W(2),
    .SKID_DEPTH_B(2),

    // Monitor parameters
    .UNIT_ID(8'h01),
    .AGENT_ID(16'h000A),
    .MAX_TRANSACTIONS(8)
) u_cg (
    .aclk(clk),
    .aresetn(rst_n),

    // Power-management interface
    .cfg_cg_enable(1'b1),           // 0 would disable the monitor
    .cfg_cg_idle_count(4'd4),       // idle cycles before the clock
    stops
    .cg_gating(cg_gating),          // high while the gated clock is
    stopped
    .cg_idle(cg_idle),

    // ... all other ports same as axil4_master_wr_mon
);
```

10.7 Verification Considerations

Because no clock is actually gated, simulation of this wrapper behaves exactly like the base module as long as `cfg_cg_enable = 1`. Drive `cfg_cg_enable = 1` for any test that expects monitor packets — with it low the monitor is off and no packets are emitted.

10.8 Related Modules

- [axil4_master_wr_mon](#) - Base module (functional specification)
- [axil4_master_rd_mon_cg](#) - Companion monitor wrapper
- [axi_monitor_base](#) - Core monitoring infrastructure
- [axi_monitor_filtered](#) - Filtering capabilities

- [AXIL4 Clock-Gated Variants Guide](#) - The transport-level _cg modules, which do perform real clock gating
-

Last Updated: 2026-07-19

10.9 Navigation

- [← Back to Base Module](#)
- [← Back to AXIL4 Index](#)
- [← Back to rtl-amba Index](#)
- [← Back to Main Documentation Index](#)

11 AXIL4 Slave Read

Module: axil4_slave_rd.sv **Location:** rtl/amba/axil4/ **Status:** ✓ Production Ready

11.1 Overview

The AXIL4 Slave Read module provides a buffered AXI4-Lite read interface for slave devices (memory, peripherals). It accepts read requests from an interconnect/master and forwards them to backend logic with elastic buffering for timing closure.

11.1.1 Key Features

- ✓ **AXI4-Lite Slave:** Buffered read-only interface (AR and R channels)
 - ✓ **Single-Beat Transactions:** No burst support
 - ✓ **Elastic Buffering:** Decouples interconnect and backend timing
 - ✓ **Activity Monitoring:** Busy signal for clock gating
 - ✓ **Minimal Latency:** 1-cycle buffer overhead per channel
-

11.2 Parameters

Parameter	Type	Default	Description
AXIL_ADDR_WIDTH	int	32	Address bus width

Parameter	Type	Default	Description
AXIL_DATA_WIDTH	int	32	Data bus width (32 or 64)
SKID_DEPTH_AR	int	2	AR channel skid buffer depth, in entries
SKID_DEPTH_R	int	4	R channel skid buffer depth, in entries

11.3 Port Groups

11.3.1 Slave Interface (Input Side from Interconnect)

Read Address Channel (AR): | Port | Direction | Width | Description | |——|———|——|
 ————| | s_axil_araddr | Input | AW | Read address (from interconnect) | |
 s_axil_arprot | Input | 3 | Protection attributes | | s_axil_arvalid | Input | 1 | Address
 valid | | s_axil_arready | Output | 1 | Address ready (buffer status) |

Read Data Channel (R): | Port | Direction | Width | Description | |——|———|——|
 ————| | s_axil_rdata | Output | DW | Read data (to interconnect) | | s_axil_rresp |
 Output | 2 | Read response | | s_axil_rvalid | Output | 1 | Data valid | | s_axil_rready |
 Input | 1 | Data ready |

11.3.2 Backend Interface (Output Side to Memory/Logic)

Read Address Channel (AR): | Port | Direction | Width | Description | |——|———|——|
 ————| | fub_araddr | Output | AW | Read address (to backend) | | fub_arprot | Output | 3
 | Protection attributes | | fub_arvalid | Output | 1 | Address valid | | fub_arready | Input | 1
 | Address ready (from backend) |

Read Data Channel (R): | Port | Direction | Width | Description | |——|———|——|
 ————| | fub_rdata | Input | DW | Read data (from backend) | | fub_rresp | Input | 2 |
 Read response | | fub_rvalid | Input | 1 | Data valid | | fub_rready | Output | 1 | Data ready
 |

11.3.3 Status Outputs

Port	Direction	Width	Description
busy	Output	1	Interface active (for

Port	Direction	Width	Description
			clock gating)

11.4 Usage Example

```

axil4_slave_rd #(
    .AXIL_ADDR_WIDTH(32),
    .AXIL_DATA_WIDTH(32),
    .SKID_DEPTH_AR(2),
    .SKID_DEPTH_R(4)
) u_axil_slave_rd (
    .aclk          (axi_clk),
    .aresetn       (axi_resetn),

    // Slave interface (from interconnect)
    .s_axil_araddr (interconnect_araddr),
    .s_axil_arprot (interconnect_arprot),
    .s_axil_arvalid (interconnect_arvalid),
    .s_axil_arready (interconnect_arready),

    .s_axil_rdata   (interconnect_rdata),
    .s_axil_rresp   (interconnect_rresp),
    .s_axil_rvalid  (interconnect_rvalid),
    .s_axil_rready  (interconnect_rready),

    // Backend interface (to memory/peripheral)
    .fub_araddr     (mem_araddr),
    .fub_arprot     (mem_arprot),
    .fub_arvalid    (mem_arvalid),
    .fub_arready    (mem_arready),

    .fub_rdata      (mem_rdata),
    .fub_rresp      (mem_rresp),
    .fub_rvalid     (mem_rvalid),
    .fub_rready     (mem_rready),

    // Status
    .busy           (rd_busy)
);

```

11.5 Design Notes

- **Signal Flow:** Interconnect → AR Buffer → Backend → R Buffer → Interconnect
 - **Use Case:** Memory controllers, peripheral slaves, register blocks
 - **Buffering:** Allows backend to respond at its own pace without stalling interconnect
-

11.6 Related Modules

- [axil4_slave_wr](#) - Slave write counterpart
 - [axil4_master_rd](#) - Master read interface
 - [axil4_slave_rd_mon](#) - Slave read with monitoring (rtl/amba/monitor/)
 - [axil4_slave_rd_cg](#) - Clock-gated version
-

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11.7 Navigation

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12 AXIL4 Slave Read Interface (Clock-Gated)

Module: `axil4_slave_rd_cg.sv` **Base Module:** [axil4_slave_rd](#) **Location:** `rtl/amba/axil4/`
Status: ✓ Production Ready

12.1 Quick Reference

This is the **clock-gated variant** of [axil4_slave_rd](#).

For complete clock-gating documentation, usage examples, and configuration guidelines, see:

→ [AXIL4 Clock-Gated Variants Guide](#) (AXIL4-specific)

12.2 Summary

The `axil4_slave_rd_cg` module adds power optimization to `axil4_slave_rd` through activity-based clock gating:

- ✓ **Same Functionality:** 100% equivalent to base module
 - ✓ **Power Savings:** dynamic clock power in the gated block scales with idle fraction; see the [AXIL4 Clock-Gated Variants Guide](#) for the estimate table and its caveats
 - ✓ **Configurable:** idle threshold and enable, both at **runtime** via `cfg_cg_idle_count` / `cfg_cg_enable`
 - ✓ **Zero Overhead When Disabled:** `cfg_cg_enable = 0` holds the clock free-running, making behavior identical to the base module
-

12.3 Common Parameters

In addition to all [axil4_slave_rd](#) parameters:

The `_cg` wrapper adds exactly **one** parameter. Gating is enabled and tuned at runtime through ports, not through parameters — there is no `ENABLE_CLOCK_GATING` parameter, no `CG_IDLE_CYCLES` parameter, and no per-domain `CG_GATE_*` parameters on this module.

Parameter	Type	Default	Description
<code>CG_IDLE_COUNT_WIDTH</code>	int	4	Width of the idle counter; max idle count = $2^N - 1$

12.3.1 Additional Ports

Port	Direction	Width	Description
<code>cfg_cg_enable</code>	Input	1	Enable clock gating (0 = clock always running)
<code>cfg_cg_idle_count</code>	Input	<code>CG_IDLE_COUNT_WIDTH</code>	Idle cycles before gating
<code>cg_gating</code>	Output	1	Clock currently gated
<code>cg_idle</code>	Output	1	No activity on the

Port	Direction	Width	Description
			previous cycle

The base module's busy output is **not** exposed on the `_cg` wrapper; it is consumed internally as a wakeup term. All other ports are identical to `axil4_slave_rd`.

12.4 Quick Usage

```
axil4_slave_rd_cg #(
    // Base module parameters (see axil4_slave_rd.md)
    .AXIL_ADDR_WIDTH(32),
    .AXIL_DATA_WIDTH(32),
    .SKID_DEPTH_AR(2),
    .SKID_DEPTH_R(4),

    // Clock gating (see CG guide for details)
    .CG_IDLE_COUNT_WIDTH(4)
) u_cg (
    .aclk(clk),
    .aresetn(rst_n),

    // Clock gating control (runtime, not parameters)
    .cfg_cg_enable(1'b1),
    .cfg_cg_idle_count(4'd5),

    // ... all other ports same as axil4_slave_rd, except `busy`,
    // which is replaced by cg_gating / cg_idle
    .cg_gating(clk_gated),
    .cg_idle(if_idle)
);
```

12.5 Documentation

- **Base Module Functionality:** [axil4_slave_rd.md](#)
- **AXIL4 Clock Gating Guide:** [axil4_clock_gating_guide.md](#)
- **Cross-Protocol Clock Gating Guide:** [clock_gated_variants.md](#)
- **Detailed CG Examples:**
 - [axi4_master_rd_mon_cg.md](#) (AXI4 monitor)
 - [axil4_master_rd_mon_cg.md](#) (AXIL4 monitor)
 - [apb4_slave_cg.md](#) (APB interface)

12.6 Navigation

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13 AXIL4 Slave Read with Monitoring

Module: axil4_slave_rd_mon.sv **Location:** rtl/amba/monitor/ **Status:** ✓ Production Ready

13.1 Overview

Combines [axil4_slave_rd](#) with [axi_monitor_filtered](#) for slave-side read monitoring.

13.1.1 Key Features

- ✓ All features of [axil4_slave_rd](#)
 - ✓ Slave-side monitoring (backend latency tracking)
 - 2-level filtering (packet-type masks, then per-event-code masks) and error detection
 - ✓ Simplified for AXI4-Lite (MAX_TRANSACTIONS=8)
-

13.2 Additional Parameters

Identical to [axil4_master_rd_mon](#) including N_ADDR_RANGES, but typically: - UNIT_ID = 2 (slaves use different unit ID) - AGENT_ID = 20 (slave agent IDs) - USE_MONITOR (synthesis-time monitor enable) - ACTIVE_TRANS_THRESHOLD (default MAX_TRANSACTIONS/2): threshold-packet trip point, replaces the former hardwired value - ENABLE_FILTERING (default 1) and ADD_PIPELINE_STAGE (default 0) - ACLK_MHZ (default 100) and CFI_MIN_FREQ_MHZ / CFI_MAX_FREQ_MHZ (default ACLK_MHZ) – the microsecond tick LUT. **Leave ACLK_MHZ at 100 on a 90 MHz part and every us-denominated timeout is wrong, silently** - USE_WDATA_ORDER_Q (default 0) and NUM_BANKS (default 1) – **NUM_BANKS > 1 on a WRITE monitor requires USE_WDATA_ORDER_Q=1**, or axi_monitor_trans_mgr fails elaboration - ADDR_FILTER_ENABLE

(default 0) – synthesises the address-range report filter; the `cfg_addr_filter_*` ports arm it at runtime

Also includes the synthesis-cone parameters `ENABLE_ERROR_LOGIC`, `ENABLE_TIMEOUT_LOGIC`, `ENABLE_COMPL_LOGIC`, `ENABLE_THRESHOLD_LOGIC`, `ENABLE_PERF_LOGIC` (all default 1) and `ENABLE_DEBUG_LOGIC` (default 0), each dropping its detection cone when set to 0. `ENABLE_PERF_PACKETS` is tied 1'b1 and `ENABLE_DEBUG_MODULE` 1'b0 internally; the removed `CAM_PIPELINE` / `TRANS_CAM_PIPELINE` parameters no longer exist.

For complete parameter descriptions including `N_ADDR_RANGES` and the synthesis-cone parameters, see [axil4_master_rd_mon](#).

13.3 Performance Monitoring

When performance monitoring is enabled, the wrapper forwards a **measurement-window state machine** plus a bank of R-channel (read-data) utilization counters to `axi_monitor_base`. All counters accumulate **only while a window is open** (`window_active = 1`) and hold their values between windows so the host can read a completed window's totals. The counters advance only while `cfg_perf_enable = 1`. `ENABLE_PERF_LOGIC = 0` does NOT drop them: the window FSM and its counters are unconditional always_ff blocks in `axi_monitor_base`, outside every generate. That parameter gates only `g_perf` in the reporter – the legacy perf-packet cone and the two lifetime counters. `USE_MONITOR = 0` is what ties the perfmon outputs off.

Avoid enabling completion (`cfg_compl_enable`) and performance (`cfg_perf_enable`) packets simultaneously under heavy traffic — the monitor bus sustains at most one packet per two cycles. Runtime-disabling either class is safe (terminal entries auto-retire; see [axi_monitor_reporter](#)); alternatively, `cfg_axi_pkt_mask` drops the packets while keeping marking and counting. See `docs/user-guides/AXI_Monitor_Configuration_Guide.md`.

13.3.1 The Measurement Window

A window is opened by a **start event** and closed by an **end event**:

- `cfg_start_event_sel` / `cfg_end_event_sel` (3-bit) select the event source (e.g. 3'b010 selects the `cfg_perf_enable` edge).
- `cfg_start_trigger` / `cfg_end_trigger` pulses fire the window directly from an engine or CSR.
- `cfg_window_force_close` is a software override that closes the window immediately.

While the window is open, `window_active` is high and `window_cycles` [31:0] free-runs, counting every clock elapsed inside the window.

13.3.2 Utilization Counters (R channel)

Every in-window cycle is classified by the R-channel valid/ready into exactly one of four buckets:

Output	Width	Condition	Meaning
<code>perf_prod_cycles</code>	32	<code>rvalid && rready</code>	productive beat transferred
<code>perf_bp_cycles</code>	32	<code>rvalid && !rready</code>	back-pressure (data offered, consumer not ready)
<code>perf_starv_cycles</code>	32	<code>!rvalid && rready</code>	starvation (consumer ready, no valid data)
<code>perf_idle_cycles</code>	32	<code>!rvalid && !rready</code>	idle

The four buckets sum to `window_cycles`, so `utilization = perf_prod_cycles / window_cycles`.

13.3.3 Throughput Counters

Output	Width	Meaning
<code>perf_beat_count</code>	32	data beats transferred (= <code>perf_prod_cycles</code> , 1 beat/cycle)
<code>perf_byte_count</code>	64	bytes transferred = beats × (1 << <code>AxSIZE</code>); AXIL is fixed at <code>AxSIZE=3'b010</code> (4 bytes)
<code>perf_burst_count</code>	32	AR (read-address) handshakes

AXI4-Lite note: every transaction is a single data beat (`ARLEN` is implicitly 0), so `perf_burst_count` counts AR handshakes = transactions and `perf_beat_count` equals the transaction count. Average burst length (`perf_beat_count / perf_burst_count`) is therefore always 1.

The `perfmon` config/status ports and the `cfg_compl_enable` / `cfg_threshold_enable` / `cfg_debug_enable` control inputs have the same directions/widths and semantics as the [read-](#)

monitor port table. As there, `cfg_debug_level/cfg_debug_mask/cfg_active_trans_threshold` are tied to constants internally and not exposed.

13.4 Monitor Backpressure (`block_ready`)

`block_ready` is an internal flow-control net inside the wrapper – it is not a port. It goes low when the monitor’s transaction-table occupancy reaches its blocking threshold (a function of `MAX_TRANSACTIONS`; the reporter FIFO depth `INTR_FIFO_DEPTH` has no path to it). The wrapper ANDs it into the upstream-facing `s_axil_arready` so a saturated monitor throttles new transactions at the handshake instead of dropping events.

- **Where the stall lands:** the upstream `s_axil_arready` is forced low until the monitor drains.
- **When `USE_MONITOR=0`:** `block_ready` is internally tied high, so the wrapper imposes no stall and runs at full bandwidth.
- **When `cfg_monitor_enable=0`:** the wrapper gate forces the upstream ready open, so a runtime-disabled monitor can never stall the datapath (the AXI4 monitors assert this as an in-RTL formal property, `ap_disabled_never_stalls`; this module has no `ifdef FORMAL` block of its own, so here the guarantee rests on the gate expression above, not a proof).

Recovery is guaranteed by the **saturation-recovery contract**: command-originated table entries are capped at `MAX_TRANSACTIONS - cmd_entry_reserve(MAX_TRANSACTIONS)` (reserve = 4 for tables of 16 or more, 0 below; the function lives in `monitor_common_pkg`), and `block_ready` re-asserts at occupancy < `MAX_TRANSACTIONS - (reserve - 1)` – a threshold strictly ABOVE the command cap – so a saturated table always drains back below the reopen point. Blocking throttles; it never deadlocks. Tables smaller than 16 keep full legacy allocation (small tables cannot spare slots) and trade the recovery guarantee for tracking capacity. The contract is verified by in-RTL formal properties (mutation-checked) and a 100-seed deliberately-undersized-table stream sweep; see [axi_monitor_base](#) for the canonical description.

Sizing note: a monitor on a bus shared by several channels/requesters must size `MAX_TRANSACTIONS` to cover `NUM_CHANNELS x per-channel outstanding` plus margin – the per-channel limit alone makes the monitor throttle the shared master. Tables deeper than 64 also need Verilator’s `--unroll-count` raised (default 64) in sim builds.

13.5 Address-Range Checker

Identical to [axil4_master_rd_mon](#) — monitors AR (read address) handshakes and emits address-range violation packets. See the master read monitor’s section for full details.

13.5.1 Packet filters

Two independent runtime filters cut monbus traffic without touching the datapath. Both are inert until driven, which is the failure to watch for: the build-time parameter only decides whether the logic is SYNTHESISED, and a design that sets the parameter but leaves the `cfg_*` ports tied low filters nothing and looks like the feature is broken.

Address-range filter — `ADDR_FILTER_ENABLE` (bit, default 0) builds it.

Port	Width	Description
<code>cfg_addr_filter_enable</code>	1	High: suppress packets for transactions outside the window. Low: inert, whatever the parameter says
<code>cfg_addr_filter_low</code>	<code>ADDR_WIDTH</code>	Window base, inclusive
<code>cfg_addr_filter_high</code>	<code>ADDR_WIDTH</code>	Window limit, inclusive

The verdict is latched per table entry at `ALLOCATION`, from the command address, and held for that entry’s life. Widening the window mid-flight does not un-filter entries already admitted — which is what makes it safe to reprogram live. Contrast the address-range `CHECKER` above, which re-evaluates per command.

There is no runtime ID filter here. AXI-Lite has no transaction IDs, so this wrapper ties `cfg_id_filter_enable` / `cfg_id_match_base` / `cfg_id_match_count` off on the inner monitor rather than exposing them — a filter keyed on a field the protocol lacks has nothing to match against.

13.6 Additional Ports

Same as [axil4_master_rd_mon](#), including the `cam_clear` control input (Input, 1) - synchronous clear of the monitor transaction CAM (driven from the harness clear control bit, e.g. `CTRL[4]`), the `cfg_compl_enable` / `cfg_threshold_enable` / `cfg_debug_enable` control enables, and the performance-monitoring config/status ports (see the [Performance Monitoring](#) section above).

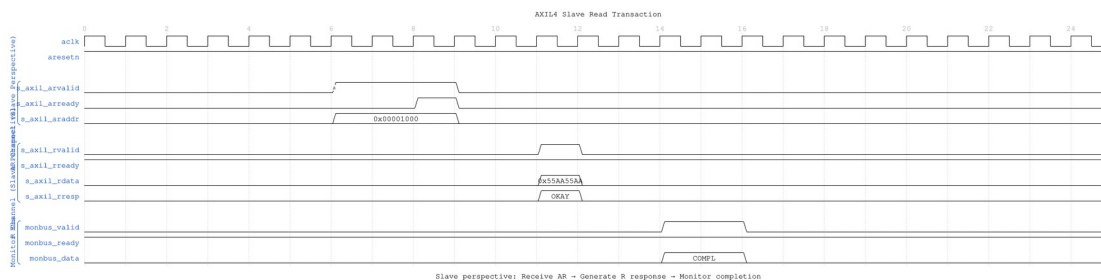
cfg_freq_sel (Input, 4) is also forwarded: the counter_freq_invariant LUT index that scales the 1 us timer tick, in which the microsecond timeouts are measured. With the default CFI_MIN_FREQ_MHZ == CFI_MAX_FREQ_MHZ == ACLK_MHZ every LUT entry is identical, so it has no effect until you give CFI a real MIN..MAX range.

13.7 Usage

```
axil4_slave_rd_mon #(
    .AXIL_ADDR_WIDTH(32),
    .AXIL_DATA_WIDTH(32),
    .UNIT_ID(2),      // Slave unit ID
    .AGENT_ID(20),    // Slave agent ID
    .MAX_TRANSACTIONS(8)
) u_axil_slave_rd_mon (
    // Slave AXIL interfaces (s_axil_*, fub_*)
    // Monitor configuration and bus
);
```

13.8 Timing Diagrams

13.8.1 Scenario 1: Slave Read Transaction

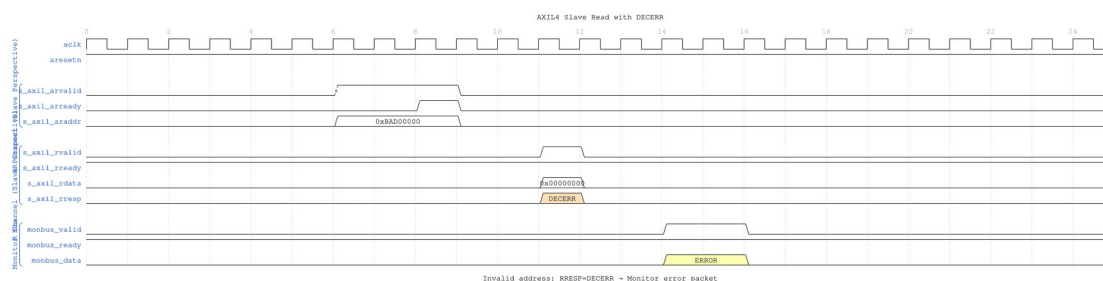


Slave Read Basic

WaveJSON: [slave_read_basic_001.json](#)

Key Observations: - Slave perspective: Receive AR request from master - Slave generates R response with data - Monitor tracks backend read latency (AR → R delay) - Completion packet indicates successful read

13.8.2 Scenario 2: Slave Read Error (DECERR)

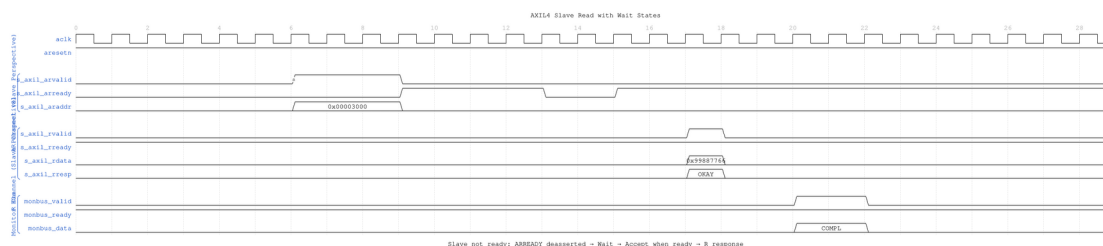


Slave Read DECERR

WaveJSON: [slave_read_decerr_001.json](#)

Key Observations: - Invalid address detected by slave address decoder - Slave returns RRESP=DECERR (decode error) - Monitor generates ERROR packet - Data value is don't-care when error occurs

13.8.3 Scenario 3: Slave Read with Wait States



Slave Read Wait

WaveJSON: [slave_read_wait_001.json](#)

Key Observations: - Slave not ready: ARREADY deasserted (wait states) - Master holds ARVALID until slave accepts - Backend read takes multiple cycles - Monitor tracks full transaction latency

13.9 Related Modules

- [axil4_slave_rd](#) - Base functional module
- [axil4_slave_wr_mon](#) - Write monitor counterpart
- [AXI4 Slave Read Mon](#) - Full AXI4 reference

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14 AXIL4 Slave Read Monitor (Clock-Gated)

Module: `axil4_slave_rd_mon_cg.sv` **Base Module:** [axil4_slave_rd_mon](#) **Location:** `rtl/amba/monitor/` **Status:** ⚠ Partial — see [Implementation Status](#)

14.1 Implementation Status

This wrapper gates the monitor’s clock for real. It instantiates `amba_clock_gate_ctrl`, and the base `axil4_slave_rd_mon` inside it is driven from `gated_aclk`, not from `aclk`.

What it does:

1. **Gates the clock.** `amba_clock_gate_ctrl` counts `cfg_cg_idle_count` idle cycles and stops `gated_aclk`, reporting on `cg_gating` (the gated clock is stopped) and `cg_idle` (no activity observed).
2. **Holds off the interfaces while gated.** `s_axil_arready` and `fub_axil_rready` are forced low under `cg_gating`, so nothing is accepted into a stopped clock.
3. **Masks the monbus valid while gated** (`monbus_valid = w_monbus_valid && !cg_gating`), which is what makes delivery exactly-once across a gating edge rather than a held valid replayed on wake.

Two earlier versions of this page were wrong in opposite directions, so both are worth naming. One described a `cg_cycles_saved` output, a `cfg_cg_idle_threshold` input and `ENABLE_CLOCK_GATING / CG_IDLE_CYCLES` parameters; none of those exist anywhere in `rtl/amba`. The other – the correction to the first – concluded that the wrapper therefore gates nothing and “will not reduce dynamic power”. That was the more damaging error: it is false, and it steers an integrator away from the right part. The real controls are `cfg_cg_enable` and `cfg_cg_idle_count` (width `CG_IDLE_COUNT_WIDTH`); there is no cycles-saved counter of any kind.

Gating behaviour is asserted directly by `val/amba/test_mon_cg_gating.py` (phase 5 for the ready hold-off, phase 6 for exactly-once monbus delivery).

14.2 Overview

`axil4_slave_rd_mon_cg` wraps [axil4_slave_rd_mon](#) and adds a power-management control and status interface. All monitoring, filtering, address-range checking, and performance-monitoring behavior is that of the base module; see [axil4_slave_rd_mon.md](#) for the complete functional specification.

14.3 Additional Parameters

In addition to all [axil4_slave_rd_mon](#) parameters (including USE_MONITOR and N_ADDR_RANGES):

Parameter	Type	Default	Description
ACLK_MHZ	int	100	Clock frequency in MHz. Builds the microsecond tick LUT in counter_freq_invariant. Leave this at 100 on a 90 MHz part and every us-denominated timeout is wrong, silently – it was unreachable through this wrapper until 2026-09-01.
CFI_MIN_FREQ_MHZ	int	ACLK_MHZ	Lowest frequency the tick LUT must cover (dynamic-frequency builds).
CFI_MAX_FREQ_MHZ	int	ACLK_MHZ	Highest frequency the tick LUT must cover.
USE_WDATA_ORDER_Q	bit	0	Write-data ordering queue. Required (=1) whenever NUM_BANKS > 1.
NUM_BANKS	int	1	Transaction-table banking. >1 needs USE_WDATA_ORDER_Q=1; the inner module's elaboration guard fires otherwise.
ADDR_FILTER_ENABLE	bit	0	Synthesises the address-range report filter. The parameter only decides

Parameter	Type	Default	Description
			whether the logic EXISTS – a build that sets it and leaves <code>cfg_addr_filter_enable</code> low filters nothing and looks broken.
<code>CG_IDLE_COUNT_WIDTH</code>	int	4	Width of <code>cfg_cg_idle_count</code> ; sets the longest programmable idle threshold

There are no `CG_GATE_MONITOR`, `CG_GATE_REPORTER`, or `CG_GATE_TIMERS` parameters, and no independent gating domains. Earlier revisions of this document described such a scheme; it was never implemented.

14.4 Additional Ports

14.4.1 Filter configuration (forwarded)

These reach the inner monitor through this wrapper; before 2026-09-01 they did not.

Port	Direction	Width	Description
<code>cfg_addr_filter_enable</code>	Input	1	High: suppress packets for transactions outside the window. Low: inert, whatever <code>ADDR_FILTER_ENABLE</code> says
<code>cfg_addr_filter_low</code>	Input	<code>ADDR_WIDTH</code>	Window base, inclusive
<code>cfg_addr_filter_high</code>	Input	<code>ADDR_WIDTH</code>	Window limit, inclusive

There is no runtime ID filter on AXI4-Lite: the protocol has no IDs to match.

All base-module ports are forwarded unchanged, including the `cam_clear` control input (Input, 1) - synchronous clear of the monitor transaction CAM (driven from the harness clear control bit, e.g. CTRL[4]). The wrapper adds:

Port	Direction	Width	Description
<code>cfg_cg_enable</code>	Input	1	Gates the monitor functionally (ANDed into <code>cfg_monitor_enable</code>). 0 = monitor disabled.

The base module's busy output remains available on this wrapper.

14.5 Performance Monitoring

The wrapper forwards the base module's full performance-monitoring interface to `axi_monitor_base` **unchanged** — the power-management interface neither adds, removes, nor retimes any perfmon port. They behave exactly as documented for [axil4_slave_rd_mon](#):

- **Config inputs:** `cfg_perf_enable`, `cfg_start_event_sel`, `cfg_end_event_sel`, `cfg_start_trigger`, `cfg_end_trigger`, `cfg_window_force_close`
- **Status / counters:** `window_active`, `window_cycles`, `perf_prod_cycles`, `perf_bp_cycles`, `perf_starv_cycles`, `perf_idle_cycles`, `perf_beat_count`, `perf_byte_count`, `perf_burst_count`

The completion/threshold/debug enables (`cfg_compl_enable`, `cfg_threshold_enable`, `cfg_debug_enable`) and the synthesis-cone parameters (`ENABLE_ERROR_LOGIC`, `ENABLE_TIMEOUT_LOGIC`, `ENABLE_COMPL_LOGIC`, `ENABLE_THRESHOLD_LOGIC`, `ENABLE_PERF_LOGIC`, `ENABLE_DEBUG_LOGIC`) are likewise forwarded unchanged. The utilization buckets watch the **R** (read-data) channel; for AXI4-Lite each transaction is a single data beat, so `perf_burst_count` counts AR handshakes = transactions.

Note that `cfg_cg_enable = 0` disables the monitor, and therefore stops the performance counters as well.

14.6 Usage Example

```
axil4_slave_rd_mon_cg #(
    // Base module parameters (see axil4_slave_rd_mon.md)
    .AXIL_ADDR_WIDTH(32),
    .AXIL_DATA_WIDTH(32),
```

```

        .SKID_DEPTH_AR(2),
        .SKID_DEPTH_R(4),

        // Monitor parameters
        .UNIT_ID(8'h01),
        .AGENT_ID(16'h000A),
        .MAX_TRANSACTIONS(8)
    ) u_cg (
        .aclk(clk),
        .aresetn(rst_n),

        // Power-management interface
        .cfg_cg_enable(1'b1),           // 0 would disable the monitor
        .cfg_cg_idle_count(4'd4),       // idle cycles before the clock
stops                                     // high while the gated clock is
stopped                                 // high while the gated clock is
        .cg_gating(cg_gating),
        .cg_idle(cg_idle),

        // ... all other ports same as axil4_slave_rd_mon
    );

```

14.7 Verification Considerations

Because no clock is actually gated, simulation of this wrapper behaves exactly like the base module as long as `cfg_cg_enable = 1`. Drive `cfg_cg_enable = 1` for any test that expects monitor packets — with it low the monitor is off and no packets are emitted.

14.8 Related Modules

- [axil4_slave_rd_mon](#) - Base module (functional specification)
 - [axil4_slave_wr_mon_cg](#) - Companion monitor wrapper
 - [axi_monitor_base](#) - Core monitoring infrastructure
 - [axi_monitor_filtered](#) - Filtering capabilities
 - [AXIL4 Clock-Gated Variants Guide](#) - The transport-level `_cg` modules, which do perform real clock gating
-

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14.9 Navigation

- [← Back to Base Module](#)
- [← Back to AXIL4 Index](#)
- [← Back to rtl-amba Index](#)
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15 AXIL4 Slave Write

Module: axil4_slave_wr.sv **Location:** rtl/amba/axil4/ **Status:** ✓ Production Ready

15.1 Overview

The AXIL4 Slave Write module provides a buffered AXI4-Lite write interface for slave devices (memory, peripherals). It accepts write requests from an interconnect/master and forwards them to backend logic with elastic buffering for timing closure.

15.1.1 Key Features

- ✓ **AXI4-Lite Slave:** Buffered write-only interface (AW, W, B channels)
 - ✓ **Single-Beat Transactions:** No burst support
 - ✓ **Byte Strobes:**WSTRB support for partial writes
 - ✓ **Elastic Buffering:** Decouples interconnect and backend timing
 - ✓ **Activity Monitoring:** Busy signal for clock gating
 - ✓ **Minimal Latency:** 1-cycle buffer overhead per channel
-

15.2 Parameters

Parameter	Type	Default	Description
AXIL_ADDR_WIDTH	int	32	Address bus width
AXIL_DATA_WIDTH	int	32	Data bus width (32 or 64)
SKID_DEPTH_AW	int	2	AW channel skid buffer

Parameter	Type	Default	Description
			depth, in entries
SKID_DEPTH_W	int	2	W channel skid buffer depth, in entries
SKID_DEPTH_B	int	2	B channel skid buffer depth, in entries

15.3 Port Groups

15.3.1 Slave Interface (Input Side from Interconnect)

Write Address Channel (AW): | Port | Direction | Width | Description | |——|———|——|
 ————| | s_axil_awaddr | Input | AW | Write address (from interconnect) | |
 s_axil_awprot | Input | 3 | Protection attributes | | s_axil_awvalid | Input | 1 | Address
 valid | | s_axil_awready | Output | 1 | Address ready (buffer status) |

Write Data Channel (W): | Port | Direction | Width | Description | |——|———|——|
 ————| | s_axil_wdata | Input | DW | Write data (from interconnect) | | s_axil_wstrb |
 Input | DW/8 | Write strobes | | s_axil_wvalid | Input | 1 | Data valid | | s_axil_wready |
 Output | 1 | Data ready (buffer status) |

Write Response Channel (B): | Port | Direction | Width | Description | |——|———|——|
 ————| | s_axil_bresp | Output | 2 | Write response (to interconnect) | | s_axil_bvalid |
 Output | 1 | Response valid | | s_axil_bready | Input | 1 | Response ready |

15.3.2 Backend Interface (Output Side to Memory/Logic)

Write Address Channel (AW): | Port | Direction | Width | Description | |——|———|——|
 ————| | fub_awaddr | Output | AW | Write address (to backend) | | fub_awprot | Output |
 3 | Protection attributes | | fub_awvalid | Output | 1 | Address valid | | fub_awready | Input |
 1 | Address ready (from backend) |

Write Data Channel (W): | Port | Direction | Width | Description | |——|———|——|
 ————| | fub_wdata | Output | DW | Write data (to backend) | | fub_wstrb | Output | DW/8
 | Write strobes | | fub_wvalid | Output | 1 | Data valid | | fub_wready | Input | 1 | Data ready
 (from backend) |

Write Response Channel (B): | Port | Direction | Width | Description | |——|———|———|
 ————| | fub_bresp | Input | 2 | Write response (from backend) | | fub_bvalid | Input | 1 |
 Response valid | | fub_bready | Output | 1 | Response ready |

15.3.3 Status Outputs

Port	Direction	Width	Description
busy	Output	1	Interface active (for clock gating)

15.4 Usage Example

```
axil4_slave_wr #(
    .AXIL_ADDR_WIDTH(32),
    .AXIL_DATA_WIDTH(32),
    .SKID_DEPTH_AW(2),
    .SKID_DEPTH_W(2),
    .SKID_DEPTH_B(2)
) u_axil_slave_wr (
    .aclk          (axi_clk),
    .aresetn       (axi_resetn),

    // Slave interface (from interconnect)
    .s_axil_awaddr (interconnect_awaddr),
    .s_axil_awprot (interconnect_awprot),
    .s_axil_awvalid (interconnect_awvalid),
    .s_axil_awready (interconnect_awready),

    .s_axil_wdata   (interconnect_wdata),
    .s_axil_wstrb   (interconnect_wstrb),
    .s_axil_wvalid  (interconnect_wvalid),
    .s_axil_wready  (interconnect_wready),

    .s_axil_bresp   (interconnect_bresp),
    .s_axil_bvalid  (interconnect_bvalid),
    .s_axil_bready  (interconnect_bready),

    // Backend interface (to memory/peripheral)
    .fub_awaddr     (mem_awaddr),
    .fub_awprot     (mem_awprot),
    .fub_awvalid    (mem_awvalid),
    .fub_awready    (mem_awready),

    .fub_wdata      (mem_wdata),
```

```

.fub_wstrb      (mem_wstrb),
.fub_wvalid     (mem_wvalid),
.fub_wready     (mem_wready),

.fub_bresp      (mem_bresp),
.fub_bvalid     (mem_bvalid),
.fub_bready     (mem_bready),

// Status
.busy           (wr_busy)
);

```

15.5 Design Notes

- **Signal Flow:** Interconnect → AW/W Buffers → Backend → B Buffer → Interconnect
 - **Use Case:** Memory controllers, peripheral slaves, register blocks
 - **Buffering:** Allows backend to respond at its own pace without stalling interconnect
 - **Write Strobes:** Backend must honor WSTRB for partial word writes
-

15.6 Related Modules

- [axil4_slave_rd](#) - Slave read counterpart
 - [axil4_master_wr](#) - Master write interface
 - [axil4_slave_wr_mon](#) - Slave write with monitoring (rtl/amba/monitor/)
 - [axil4_slave_wr_cg](#) - Clock-gated version
-

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15.7 Navigation

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16 AXIL4 Slave Write Interface (Clock-Gated)

Module: `axil4_slave_wr_cg.sv` **Base Module:** [axil4_slave_wr](#) **Location:** `rtl/amba/axil4/`
Status: ✓ Production Ready

16.1 Quick Reference

This is the **clock-gated variant** of [axil4_slave_wr](#).

For complete clock-gating documentation, usage examples, and configuration guidelines, see:

- [AXIL4 Clock-Gated Variants Guide](#) (AXIL4-specific)
 - [Clock-Gated Variants Guide](#) (cross-protocol overview)
-

16.2 Summary

The `axil4_slave_wr_cg` module adds power optimization to `axil4_slave_wr` through activity-based clock gating:

- ✓ **Same Functionality:** 100% equivalent to base module
 - ✓ **Power Savings:** dynamic clock power in the gated block scales with idle fraction; see the [AXIL4 Clock-Gated Variants Guide](#) for the estimate table and its caveats
 - ✓ **Configurable:** idle threshold and enable, both at **runtime** via `cfg_cg_idle_count` / `cfg_cg_enable`
 - ✓ **Zero Overhead When Disabled:** `cfg_cg_enable = 0` holds the clock free-running, making behavior identical to the base module
-

16.3 Common Parameters

In addition to all [axil4_slave_wr](#) parameters:

The `_cg` wrapper adds exactly **one** parameter. Gating is enabled and tuned at runtime through ports, not through parameters — there is no `ENABLE_CLOCK_GATING` parameter, no `CG_IDLE_CYCLES` parameter, and no per-domain `CG_GATE_*` parameters on this module.

Parameter	Type	Default	Description
CG_IDLE_COUNT_WIDTH	int	4	Width of the idle counter; max idle count = $2^N - 1$

16.3.1 Additional Ports

Port	Direction	Width	Description
cfg_cg_enable	Input	1	Enable clock gating (0 = clock always running)
cfg_cg_idle_count	Input	CG_IDLE_COUNT_WIDTH	Idle cycles before gating
cg_gating	Output	1	Clock currently gated
cg_idle	Output	1	No activity on the previous cycle

The base module's busy output is **not** exposed on the _cg wrapper; it is consumed internally as a wakeup term. All other ports are identical to axil4_slave_wr.

16.4 Quick Usage

```
axil4_slave_wr_cg #(
    // Base module parameters (see axil4_slave_wr.md)
    .AXIL_ADDR_WIDTH(32),
    .AXIL_DATA_WIDTH(32),
    .SKID_DEPTH_AW(2),
    .SKID_DEPTH_W(2),
    .SKID_DEPTH_B(2),

    // Clock gating (see CG guide for details)
    .CG_IDLE_COUNT_WIDTH(4)
) u_cg (
    .aclk(clk),
    .aresetn(rst_n),

    // Clock gating control (runtime, not parameters)
    .cfg_cg_enable(1'b1),
    .cfg_cg_idle_count(4'd5),

    // ... all other ports same as axil4_slave_wr, except `busy`,
    // which is replaced by cg_gating / cg_idle
```

```
.cg_gating(clk_gated),  
.cg_idle(if_idle)  
);
```

16.5 Documentation

- **Base Module Functionality:** [axil4_slave_wr.md](#)
 - **AXIL4 Clock Gating Guide:** [axil4_clock_gating_guide.md](#)
 - **Cross-Protocol Clock Gating Guide:** [clock_gated_variants.md](#)
 - **Detailed CG Examples:**
 - [axi4_master_rd_mon_cg.md](#) (AXI4 monitor)
 - [axil4_master_rd_mon_cg.md](#) (AXIL4 monitor)
 - [apb4_slave_cg.md](#) (APB interface)
-

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16.6 Navigation

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- [← Back to Main Documentation Index](#)

17 AXIL4 Slave Write with Monitoring

Module: `axil4_slave_wr_mon.sv` **Location:** `rtl/amba/monitor/` **Status:** ✓ Production Ready

17.1 Overview

Combines [axil4_slave_wr](#) with `axi_monitor_filtered` for slave-side write monitoring.

17.1.1 Key Features

- ✓ All features of `axil4_slave_wr`
- ✓ Slave-side write monitoring (AW, W, B channels)
- ✓ Backend write latency tracking

- 2-level filtering (packet-type masks, then per-event-code masks) and error detection
-

17.2 Additional Parameters

- `UNIT_ID` = 2 (slaves)
- `AGENT_ID` = 21 (slave write agent)
- `USE_MONITOR` (synthesis-time monitor enable)
- `ACTIVE_TRANS_THRESHOLD` (default `MAX_TRANSACTIONS/2`): threshold-packet trip point, replaces the former hardwired value
- `ENABLE_FILTERING` (default 1) and `ADD_PIPELINE_STAGE` (default 0)
- `ACLK_MHZ` (default 100) and `CFI_MIN_FREQ_MHZ` / `CFI_MAX_FREQ_MHZ` (default `ACLK_MHZ`) – the microsecond tick LUT. **Leave `ACLK_MHZ` at 100 on a 90 MHz part and every us-denominated timeout is wrong, silently**
- `USE_WDATA_ORDER_Q` (default 0) and `NUM_BANKS` (default 1) – **`NUM_BANKS > 1` on a WRITE monitor requires `USE_WDATA_ORDER_Q=1`**, or `axi_monitor_trans_mgr` fails elaboration
- `ADDR_FILTER_ENABLE` (default 0) – synthesises the address-range report filter; the `cfg_addr_filter_*` ports arm it at runtime
- `N_ADDR_RANGES` (address-range comparator count)
- Others same as [axil4_master_rd_mon](#)

Also includes the synthesis-cone parameters `ENABLE_ERROR_LOGIC`, `ENABLE_TIMEOUT_LOGIC`, `ENABLE_COMPL_LOGIC`, `ENABLE_THRESHOLD_LOGIC`, `ENABLE_PERF_LOGIC` (all default 1) and `ENABLE_DEBUG_LOGIC` (default 0), each dropping its detection cone when set to 0. `ENABLE_PERF_PACKETS` is tied 1'b1 and `ENABLE_DEBUG_MODULE` 1'b0 internally; the removed `CAM_PIPELINE` / `TRANS_CAM_PIPELINE` parameters no longer exist.

For complete parameter descriptions, see [axil4_master_rd_mon](#).

17.3 Performance Monitoring

When performance monitoring is enabled, the wrapper forwards a **measurement-window state machine** plus a bank of W-channel (write-data) utilization counters to `axi_monitor_base`. All counters accumulate **only while a window is open** (`window_active` = 1) and hold their values between windows so the host can read a completed window's totals. The counters advance only while `cfg_perf_enable` = 1. `ENABLE_PERF_LOGIC` = 0 does NOT drop them: the window FSM and its counters are unconditional always_ff blocks in `axi_monitor_base`, outside every

generate. That parameter gates only `g_perf` in the reporter – the legacy perf-packet cone and the two lifetime counters. `USE_MONITOR = 0` is what ties the perfmon outputs off.

Avoid enabling completion (`cfg_compl_enable`) and performance (`cfg_perf_enable`) packets simultaneously under heavy traffic — the monitor bus sustains at most one packet per two cycles. Runtime-disabling either class is safe (terminal entries auto-retire; see [axi_monitor_reporter](#)); alternatively, `cfg_axi_pkt_mask` drops the packets while keeping marking and counting. See `docs/user-guides/AXI_Monitor_Configuration_Guide.md`.

17.3.1 The Measurement Window

A window is opened by a **start event** and closed by an **end event**:

- `cfg_start_event_sel / cfg_end_event_sel` (3-bit) select the event source (e.g. `3'b010` selects the `cfg_perf_enable` edge).
- `cfg_start_trigger / cfg_end_trigger` pulses fire the window directly from an engine or CSR.
- `cfg_window_force_close` is a software override that closes the window immediately.

While the window is open, `window_active` is high and `window_cycles` [31:0] free-runs, counting every clock elapsed inside the window.

17.3.2 Utilization Counters (W channel)

Every in-window cycle is classified by the W-channel valid/ready into exactly one of four buckets:

Output	Width	Condition	Meaning
<code>perf_prod_cycles</code>	32	<code>wvalid && wready</code>	productive beat transferred
<code>perf_bp_cycles</code>	32	<code>wvalid && !wready</code>	back-pressure (data offered, consumer not ready)
<code>perf_starv_cycles</code>	32	<code>!wvalid && wready</code>	starvation (consumer ready, no valid data)
<code>perf_idle_cycles</code>	32	<code>!wvalid && !wready</code>	idle

The four buckets sum to `window_cycles`, so $\text{utilization} = \text{perf_prod_cycles} / \text{window_cycles}$.

17.3.3 Throughput Counters

Output	Width	Meaning
<code>perf_beat_count</code>	32	data beats transferred (= <code>perf_prod_cycles</code> , 1 beat/cycle)
<code>perf_byte_count</code>	64	bytes transferred = beats × (1 << <code>AxSIZE</code>); AXIL is fixed at <code>AxSIZE=3'b010</code> (4 bytes)
<code>perf_burst_count</code>	32	AW (write-address) handshakes

AXI4-Lite note: every transaction is a single data beat (`AWLEN` is implicitly 0), so `perf_burst_count` counts AW handshakes = transactions and `perf_beat_count` equals the transaction count. Average burst length (`perf_beat_count / perf_burst_count`) is therefore always 1.

The `perfmon` config/status ports and the `cfg_compl_enable / cfg_threshold_enable / cfg_debug_enable` control inputs have the same directions/widths and semantics as the [read-monitor port table](#) (only `perf_burst_count` counts AW handshakes here). As there, `cfg_debug_level/cfg_debug_mask/cfg_active_trans_threshold` are tied to constants internally and not exposed.

17.4 Monitor Backpressure (`block_ready`)

`block_ready` is an internal flow-control net inside the wrapper – it is not a port. It goes low when the monitor’s transaction-table occupancy reaches its blocking threshold (a function of `MAX_TRANSACTIONS`; the reporter FIFO depth `INTR_FIFO_DEPTH` has no path to it). The wrapper ANDs it into the upstream-facing `s_axil_awready` so a saturated monitor throttles new transactions at the handshake instead of dropping events.

- **Where the stall lands:** the upstream `s_axil_awready` is forced low until the monitor drains.
- **When `USE_MONITOR=0`:** `block_ready` is internally tied high, so the wrapper imposes no stall and runs at full bandwidth.
- **When `cfg_monitor_enable=0`:** the wrapper gate forces the upstream ready open, so a runtime-disabled monitor can never stall the datapath (the AXI4

monitors assert this as an in-RTL formal property, `ap_disabled_never_stalls`; this module has no `ifdef FORMAL` block of its own, so here the guarantee rests on the gate expression above, not a proof).

Recovery is guaranteed by the **saturation-recovery contract**: command-originated table entries are capped at `MAX_TRANSACTIONS - cmd_entry_reserve(MAX_TRANSACTIONS)` (reserve = 4 for tables of 16 or more, 0 below; the function lives in `monitor_common_pkg`), and `block_ready` re-asserts at occupancy < `MAX_TRANSACTIONS - (reserve - 1)` – a threshold strictly ABOVE the command cap – so a saturated table always drains back below the reopen point. Blocking throttles; it never deadlocks. Tables smaller than 16 keep full legacy allocation (small tables cannot spare slots) and trade the recovery guarantee for tracking capacity. The contract is verified by in-RTL formal properties (mutation-checked) and a 100-seed deliberately-undersized-table stream sweep; see [axi_monitor_base](#) for the canonical description.

Sizing note: a monitor on a bus shared by several channels/requesters must size `MAX_TRANSACTIONS` to cover `NUM_CHANNELS × per-channel outstanding` plus margin – the per-channel limit alone makes the monitor throttle the shared master. Tables deeper than 64 also need Verilator's `--unroll-count` raised (default 64) in sim builds.

17.5 Address-Range Checker

Identical to [axil4_master_rd_mon](#) except the checker watches AW (write address) handshakes. The `cfg_addr_*` configuration inputs and monbus event encoding are the same. See the read monitor's Address-Range Checker section for full details.

17.5.1 Packet filters

Two independent runtime filters cut monbus traffic without touching the datapath. Both are inert until driven, which is the failure to watch for: the build-time parameter only decides whether the logic is SYNTHESISED, and a design that sets the parameter but leaves the `cfg_*` ports tied low filters nothing and looks like the feature is broken.

Address-range filter — `ADDR_FILTER_ENABLE` (bit, default 0) builds it.

Port	Width	Description
<code>cfg_addr_filter_enable</code>	1	High: suppress packets for transactions outside the window. Low: inert, whatever the parameter says

Port	Width	Description
cfg_addr_filter_low	ADDR_WIDTH	Window base, inclusive
cfg_addr_filter_high	ADDR_WIDTH	Window limit, inclusive

The verdict is latched per table entry at ALLOCATION, from the command address, and held for that entry's life. Widening the window mid-flight does not un-filter entries already admitted — which is what makes it safe to reprogram live. Contrast the address-range CHECKER above, which re-evaluates per command.

There is no runtime ID filter here. AXI-Lite has no transaction IDs, so this wrapper ties `cfg_id_filter_enable` / `cfg_id_match_base` / `cfg_id_match_count` off on the inner monitor rather than exposing them — a filter keyed on a field the protocol lacks has nothing to match against.

17.6 Additional Ports

Same as [axil4_master_rd_mon](#), including the `cam_clear` control input (Input, 1) - synchronous clear of the monitor transaction CAM (driven from the harness clear control bit, e.g. CTRL[4]), the `cfg_compl_enable` / `cfg_threshold_enable` / `cfg_debug_enable` control enables, and the performance-monitoring config/status ports (see the [Performance Monitoring](#) section above).

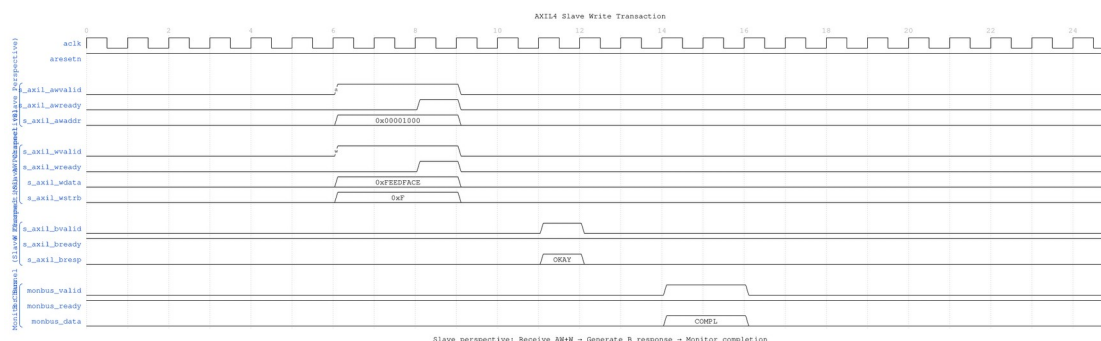
`cfg_freq_sel` (Input, 4) is also forwarded: the `counter_freq_invariant` LUT index that scales the 1 us timer tick, in which the microsecond timeouts are measured. With the default `CFI_MIN_FREQ_MHZ == CFI_MAX_FREQ_MHZ == ACLK_MHZ` every LUT entry is identical, so it has no effect until you give CFI a real MIN..MAX range.

17.7 Usage

```
axil4_slave_wr_mon #(
    .AXIL_ADDR_WIDTH(32),
    .AXIL_DATA_WIDTH(32),
    .UNIT_ID(2),
    .AGENT_ID(21),
    .MAX_TRANSACTIONS(8)
) u_axil_slave_wr_mon (
    // Slave AXIL write interfaces
    // Monitor configuration and bus
);
```

17.8 Timing Diagrams

17.8.1 Scenario 1: Slave Write Transaction

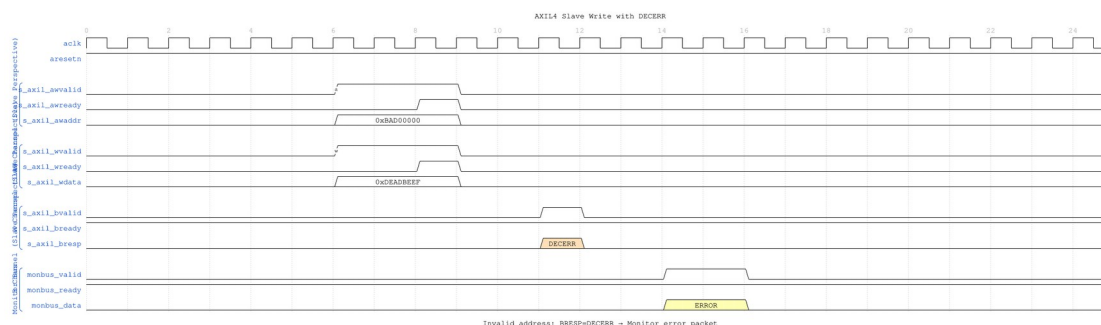


Slave Write Basic

WaveJSON: [slave_write_basic_001.json](#)

Key Observations: - Slave perspective: Receive AW+W simultaneously - Slave commits write to backend storage - Slave generates B response with BRESP=OKAY - Monitor tracks backend write latency

17.8.2 Scenario 2: Slave Write Error (DECERR)

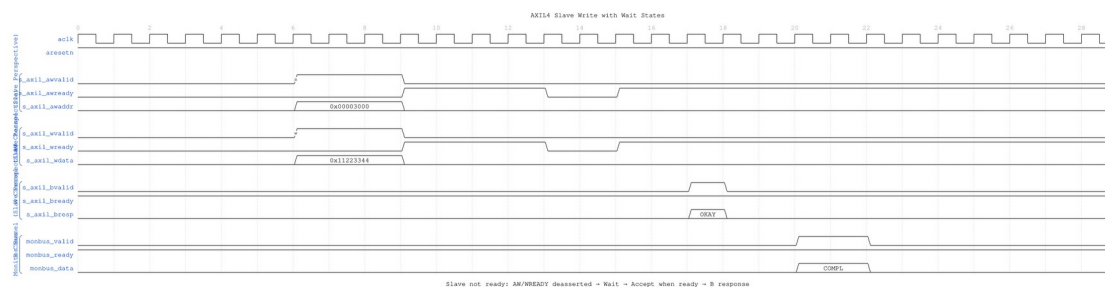


Slave Write DECERR

WaveJSON: [slave_write_decerr_001.json](#)

Key Observations: - Invalid address detected by slave - Write data received but not committed - Slave returns BRESP=DECERR - Monitor generates ERROR packet

17.8.3 Scenario 3: Slave Write with Wait States



Slave Write Wait

WaveJSON: [slave_write_wait_001.json](#)

Key Observations: - Slave not ready: AWREADY/WREADY deasserted - Master holds AW+W until slave accepts - Backend write processing delay - Monitor tracks full write latency including wait states

17.9 Related Modules

- [axil4_slave_wr](#) - Base functional module
- [axil4_slave_rd_mon](#) - Read monitor counterpart
- [AXI4 Slave Write Mon](#) - Full AXI4 reference

Last Updated: 2026-07-19

18 AXIL4 Slave Write Monitor (Clock-Gated)

Module: axil4_slave_wr_mon_cg.sv **Base Module:** [axil4_slave_wr_mon](#) **Location:** rtl/amba/monitor/ **Status:** ⚠ Partial — see [Implementation Status](#)

18.1 Implementation Status

This wrapper gates the monitor's clock for real. It instantiates `amba_clock_gate_ctrl`, and the base `axil4_slave_wr_mon` inside it is driven from `gated_aclk`, not from `aclk`.

What it does:

1. **Gates the clock.** `amba_clock_gate_ctrl` counts `cfg_cg_idle_count` idle cycles and stops `gated_aclk`, reporting on `cg_gating` (the gated clock is stopped) and `cg_idle` (no activity observed).
2. **Holds off the interfaces while gated.** `s_axil_awready`, `s_axil_wready` and `fub_axil_bready` are forced low under `cg_gating`, so nothing is accepted into a stopped clock.
3. **Masks the monbus valid while gated** (`monbus_valid = w_monbus_valid && !cg_gating`), which is what makes delivery exactly-once across a gating edge rather than a held valid replayed on wake.

Two earlier versions of this page were wrong in opposite directions, so both are worth naming. One described a `cg_cycles_saved` output, a `cfg_cg_idle_threshold` input and `ENABLE_CLOCK_GATING / CG_IDLE_CYCLES` parameters; none of those exist anywhere in `rtl/amba`. The other – the correction to the first – concluded that the wrapper therefore gates nothing and “will not reduce dynamic power”. That was the more damaging error: it is false, and it steers an integrator away from the right part. The real controls are `cfg_cg_enable` and `cfg_cg_idle_count` (width `CG_IDLE_COUNT_WIDTH`); there is no cycles-saved counter of any kind.

Gating behaviour is asserted directly by `val/amba/test_mon_cg_gating.py` (phase 5 for the ready hold-off, phase 6 for exactly-once monbus delivery).

18.2 Overview

`axil4_slave_wr_mon_cg` wraps [axil4_slave_wr_mon](#) and adds a power-management control and status interface. All monitoring, filtering, address-range checking, and performance-monitoring behavior is that of the base module; see [axil4_slave_wr_mon.md](#) for the complete functional specification.

18.3 Additional Parameters

In addition to all [axil4_slave_wr_mon](#) parameters (including `USE_MONITOR` and `N_ADDR_RANGES`):

Parameter	Type	Default	Description
<code>ACLK_MHZ</code>	<code>int</code>	100	Clock frequency in MHz. Builds the microsecond tick LUT in <code>counter_freq_invarian</code>

Parameter	Type	Default	Description
			t. Leave this at 100 on a 90 MHz part and every us-denominated timeout is wrong, silently – it was unreachable through this wrapper until 2026-09-01.
CFI_MIN_FREQ_MHZ	int	ACLK_MHZ	Lowest frequency the tick LUT must cover (dynamic-frequency builds).
CFI_MAX_FREQ_MHZ	int	ACLK_MHZ	Highest frequency the tick LUT must cover.
USE_WDATA_ORDER_Q	bit	0	Write-data ordering queue. Required (=1) whenever NUM_BANKS > 1.
NUM_BANKS	int	1	Transaction-table banking. >1 needs USE_WDATA_ORDER_Q=1; the inner module's elaboration guard fires otherwise.
ADDR_FILTER_ENABLE	bit	0	Synthesises the address-range report filter. The parameter only decides whether the logic EXISTS – a build that sets it and leaves cfg_addr_filter_enable low filters nothing and looks broken.
CG_IDLE_COUNT_WIDTH	int	4	Width of cfg_cg_idle_count; sets the longest

Parameter	Type	Default	Description
			programmable idle threshold

There are no CG_GATE_MONITOR, CG_GATE_REPORTER, or CG_GATE_TIMERS parameters, and no independent gating domains. Earlier revisions of this document described such a scheme; it was never implemented.

18.4 Additional Ports

18.4.1 Filter configuration (forwarded)

These reach the inner monitor through this wrapper; before 2026-09-01 they did not.

Port	Direction	Width	Description
cfg_addr_filter_enable	Input	1	High: suppress packets for transactions outside the window. Low: inert, whatever ADDR_FILTER_ENABLE says
cfg_addr_filter_low	Input	ADDR_WIDTH	Window base, inclusive
cfg_addr_filter_high	Input	ADDR_WIDTH	Window limit, inclusive

There is no runtime ID filter on AXI4-Lite: the protocol has no IDs to match.

All base-module ports are forwarded unchanged, including the cam_clear control input (Input, 1) - synchronous clear of the monitor transaction CAM (driven from the harness clear control bit, e.g. CTRL[4]). The wrapper adds:

Port	Direction	Width	Description
cfg_cg_enable	Input	1	Gates the monitor functionally (ANDed into cfg_monitor_enable). 0 = monitor disabled.

The base module's busy output remains available on this wrapper.

18.5 Performance Monitoring

The wrapper forwards the base module's full performance-monitoring interface to `axi_monitor_base` **unchanged** — the power-management interface neither adds, removes, nor retimes any perfmon port. They behave exactly as documented for [axil4_slave_wr_mon](#):

- **Config inputs:** `cfg_perf_enable`, `cfg_start_event_sel`, `cfg_end_event_sel`, `cfg_start_trigger`, `cfg_end_trigger`, `cfg_window_force_close`
- **Status / counters:** `window_active`, `window_cycles`, `perf_prod_cycles`, `perf_bp_cycles`, `perf_starv_cycles`, `perf_idle_cycles`, `perf_beat_count`, `perf_byte_count`, `perf_burst_count`

The completion/threshold/debug enables (`cfg_compl_enable`, `cfg_threshold_enable`, `cfg_debug_enable`) and the synthesis-cone parameters (`ENABLE_ERROR_LOGIC`, `ENABLE_TIMEOUT_LOGIC`, `ENABLE_COMPL_LOGIC`, `ENABLE_THRESHOLD_LOGIC`, `ENABLE_PERF_LOGIC`, `ENABLE_DEBUG_LOGIC`) are likewise forwarded unchanged. The utilization buckets watch the **W** (write-data) channel; for AXI4-Lite each transaction is a single data beat, so `perf_burst_count` counts AW handshakes = transactions.

Note that `cfg_cg_enable = 0` disables the monitor, and therefore stops the performance counters as well.

18.6 Usage Example

```
axil4_slave_wr_mon_cg #(
    // Base module parameters (see axil4_slave_wr_mon.md)
    .AXIL_ADDR_WIDTH(32),
    .AXIL_DATA_WIDTH(32),
    .SKID_DEPTH_AW(2),
    .SKID_DEPTH_W(2),
    .SKID_DEPTH_B(2),

    // Monitor parameters
    .UNIT_ID(8'h01),
    .AGENT_ID(16'h000A),
    .MAX_TRANSACTIONS(8)
) u_cg (
    .aclk(clk),
    .aresetn(rst_n),

    // Power-management interface
    .cfg_cg_enable(1'b1),           // 0 would disable the monitor
    .cfg_cg_idle_count(4'd4),       // idle cycles before the clock
```

```
stops
    .cg_gating(cg_gating),           // high while the gated clock is
stopped
    .cg_idle(cg_idle),

    // ... all other ports same as axil4_slave_wr_mon
);
```

18.7 Verification Considerations

Because no clock is actually gated, simulation of this wrapper behaves exactly like the base module as long as `cfg_cg_enable = 1`. Drive `cfg_cg_enable = 1` for any test that expects monitor packets — with it low the monitor is off and no packets are emitted.

18.8 Related Modules

- [axil4_slave_wr_mon](#) - Base module (functional specification)
 - [axil4_slave_rd_mon_cg](#) - Companion monitor wrapper
 - [axi_monitor_base](#) - Core monitoring infrastructure
 - [axi_monitor_filtered](#) - Filtering capabilities
 - [AXIL4 Clock-Gated Variants Guide](#) - The transport-level `_cg` modules, which do perform real clock gating
-

Last Updated: 2026-07-19

18.9 Navigation

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